Is correlation meaningfully correlated with causation? Crud, 1/f spectra, and an empirical null for small-effect observational claims

Significance Statement

Observational science routinely interprets small adjusted correlations as evidence for causal effects. We test this assumption in nine large datasets and find that most pairs of measured variables are correlated to a non-trivial degree, even after standard statistical adjustments. As a consequence, many associations that look "significant" by classical standards are not distinctive: their magnitudes are typical of randomly chosen variable pairs in the same dataset. We propose a diagnostic—a crud-aware test—that benchmarks any reported correlation against the empirical null of same-pipeline residuals. Only associations that clearly exceed this background should be treated as evidence for a direct causal relationship; small adjusted correlations should be reported with the caveat that their magnitude alone is not a basis for causal claims.

Abstract

Causal claims in observational research typically start with two measured variables, adjust for confounders, and interpret the remaining correlation as evidence of a causal effect. For this logic to work, adjusted correlations must be informative about their underlying causal effects. We test this assumption using nine large datasets spanning health, neuroscience, psychology, genomics, politics, and vision. Across these datasets, most pairwise correlations are classically significant—the so-called "crud" background. We trace this background to approximately $1/f$ eigenvalue spectra, which distribute shared variance across many directions so that the residual crud scale σ_K shrinks only slowly with the number of removed components: within the fixed-K regime we model, the moderate-dimensional adjustments we tested (PC regression and Lasso) do not reduce σ_K to negligible levels in any of the nine datasets, regardless of sample size. A decision-theoretic benchmark formalizes the consequence: under a stylized Gaussian mixture, the Bayes error for magnitude-only classification (any rule using only the adjusted-association statistic) is bounded below by a constant when the causal signal is comparable to σ_K. We propose a crud-aware diagnostic that benchmarks observed correlations against the domain's empirical null—a percentile rank in the empirical version, a z-test in a parametric small-study version. In most domains studied here, even a correlation as large as $r = 0.10$ does not clear the crud-aware threshold.

1 Introduction

Across biomedicine, social science, and behavioral research, causal claims increasingly rely on observational associations in large datasets. Randomized experiments remain the benchmark for counterfactual effects, but intervention is often infeasible, so small adjusted associations are routinely interpreted as evidence about causal mechanism. For this logic to work, adjusted correlations must be informative about their underlying causal effects. We test that assumption by asking how

informative those associations remain in the presence of the background dependence structure typical of modern datasets. We focus on small-effect causal claims that depend on adjusted associations standing out from background dependence; large effects easily clear any reasonable background, and prediction tasks do not require causal interpretation.

Related work emphasizes that widespread weak dependence is common: Meehl's "crud factor" [Meehl, 1990] argued that in large behavioral datasets, most pairs of variables are non-trivially correlated; Branwen [n.d.] reviews the "everything is correlated" perspective. Sensitivity analyses [Cinelli and Hazlett, 2020, Oster, 2019] ask the hypothetical question of how strong unobserved confounding would need to be to nullify a result; we ask the empirical question of what background dependence actually looks like in a domain. Approximate $1/f$ spectral structure has been documented across many complex systems [Press, 1978, Keshner, 1982, Bak et al., 1987]. Our contribution connects these threads: we quantify the background correlation distribution across domains, link its slow shrinkage under generic adjustment to eigenvalue spectra, and propose a crud-aware test for interpreting small adjusted associations. A complementary literature documents that small published effects are often inflated [Button et al., 2013, Szucs and Ioannidis, 2017, Open Science Collaboration, 2015, Gelman and Carlin, 2014, Ioannidis, 2005, 2008]; SI Appendix F surveys this evidence field by field and shows that the typical published effect would not clear our test in observational medicine, nutrition, common-variant genetics, or BWAS-style neuroscience.

A concrete example: a recent study reported a nominally significant association between folate intake and mortality in adults with type-2 diabetes ($r \approx 0.02$, $p_{\text{classical}} < 0.05$, $n \approx 8{,}000$) [Liu et al., 2022]. Compared to the background distribution of correlations among NHANES variables, however, it falls in the middle: the crud-aware p-value is 0.81 (Section 2.8). The classical test detects that the correlation is nonzero; it does not detect that the correlation is distinctive.

We operationalize the background dependence scale, for this paper, as the residual correlation standard deviation across uniformly sampled feature pairs after a fixed adjustment (regression on the top K principal components), and use it as the benchmark for whether an adjusted association is surprising. Whether this matches a domain's conceptual notion of background depends on whether the adjustment captures the domain's confounding structure. The empirical null is not a "no direct edge" null—in real domains, pervasive confounding ensures that uniformly sampled pairs share indirect dependence—it is the distribution of typical same-pipeline residual correlations. The argument is correspondingly weaker: a target sitting in the bulk cannot be cleanly read as evidence of a direct causal relation, because the same residual magnitude can arise from typical residual confounding alone. Across our nine domains, most small adjusted associations do not separate from the empirical null. We (i) document this empirically, (ii) trace it to approximately power-law eigenvalue spectra, (iii) prove a Bayes-error benchmark (Theorems 2–3 in SI Appendix D) for magnitude-only classification under a stylized mixture, and (iv) propose a crud-aware test against the empirical null.

2 Results

2.1 Datasets

We analyze nine large datasets spanning neuroscience, genomics, psychology, political science, vision, and population health (Figure 1; Methods; SI Appendix A gives full descriptions and provenance): NHANES, precinct-level 2016 voting and Census demographics, GTEx v8 RNA-seq, Allen Institute mouse brain RNA-seq, HEXACO, CIFAR-10, Kay fMRI, Haxby fMRI, and Stringer mouse V1. The nine span dimensionalities (p from 83 to 6,000), raw sample sizes (n from ∼8,000 to

∼30,000; analysis subsamples to $n = 10{,}000$ where larger, see Methods), and measurement types (continuous neural signals, binary/ordinal survey items, count-based expression, pixel intensities), so patterns common to all cannot easily be attributed to a single data modality. This is a convenience sample of large public datasets; generalization beyond the nine remains an empirical question.

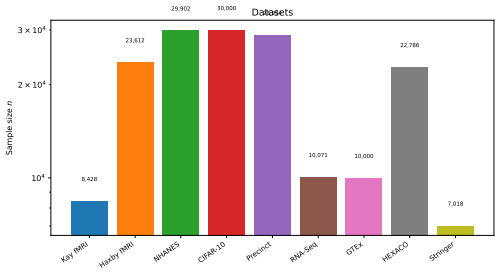

Figure 1: Nine large datasets used to study correlational structure across modern science domains.

2.2 Empirical finding: broad correlations persist after adjustment

For uniformly sampled feature pairs, we observe a broad spread of correlations (Figure 2, left; Table 1). Given the large sample sizes, classical iid correlation tests label most of these correlations as nominally significant, even though most random pairs are implausible candidates for direct causal relations. After PC regression at $K = 10$ (Methods; sensitivity across K in Table 1), the residual correlation distribution remains wide (Figure 2, right). The core empirical fact: the background correlation distribution is wide even after generic adjustment, so "significant" does not mean "distinctive." This is not a multiple-testing point: even ignoring p-values, the typical residual correlation scale after generic adjustment is substantial, so small adjusted associations are not distinctive evidence in the first place. Across the low-rank adjustments we tested (PC regression and Lasso, SI Appendix G.4), none collapsed the residual distribution on any of the nine datasets.

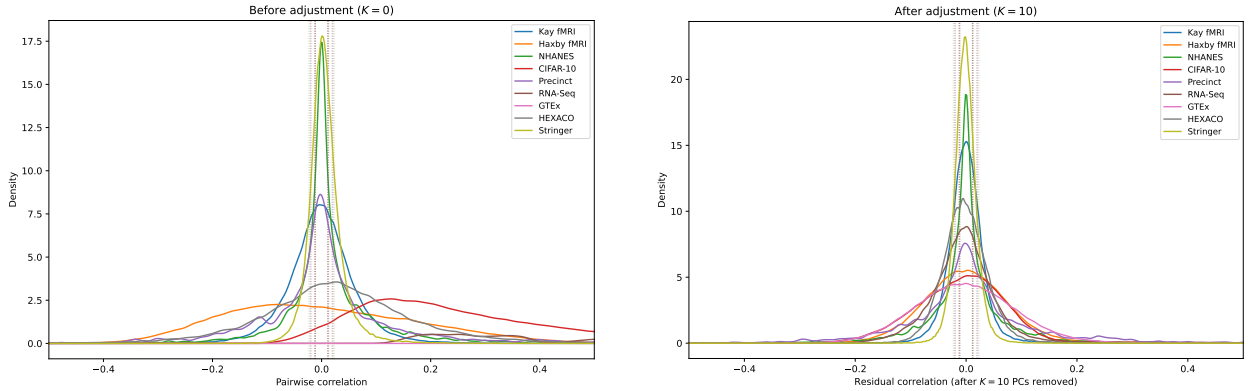

Figure 2: **Broad distributions of correlations persist after generic low-rank adjustment.** Left: distribution of pairwise correlations across uniformly sampled feature pairs before adjustment ($K = 0$). Right: the same distribution after regressing each feature on the top $K = 10$ principal component scores. Dotted vertical lines mark the classical two-sided significance threshold ($p < 0.05$) for each dataset's full sample size; nearly the entire empirical null exceeds this threshold.

Table 1: Off-diagonal correlation spread (SD) after removing the top K principal components. Entries report the standard deviation of off-diagonal correlations across uniformly sampled feature pairs. The spread generally decreases with K but remains nontrivial in all datasets.

Dataset	$K = 0$	$K = 1$	$K = 10$	$K = 50$
Kay fMRI	0.065	0.055	0.035	0.029
Haxby fMRI	0.175	0.150	0.077	0.038
NHANES	0.115	0.102	0.093	0.117
CIFAR-10	0.191	0.212	0.105	0.064
Precinct	0.166	0.159	0.136	0.293†
RNA-Seq	0.170	0.196	0.071	0.043
GTEx	0.038	0.255	0.090	0.043
HEXACO	0.144	0.125	0.046	0.040
Stringer	0.035	0.031	0.021	0.021

†Precinct has only p=83 features; removing 50 of 83 PCs leaves a near-degenerate residual subspace.

To give these numbers a concrete interpretation, for $r = 0.10$ to clear the conventional 95th-percentile crud-aware threshold ($z_{\text{crud}} > 1.96$, where $z_{\text{crud}} = |r|/\sqrt{\sigma_K^2 + 1/(n-1)}$) the crud scale must satisfy $\sigma_K \lesssim 0.05$. At $K = 0$ this holds in only two of nine datasets (Stringer, GTEx); after removing ten PCs, only three (Stringer, Kay, HEXACO). In the remaining six datasets, $r = 0.10$ does not clear the threshold at $K = 10$ within the fixed-K model, regardless of n.

2.3 Mechanism: power-law spectra

Why does removing broad shared variation not shrink the correlation distribution faster? Across all nine datasets, explained variance decays approximately as a power law (Figure 3): the spectrum is roughly linear on log–log axes, meaning shared variation is distributed across many components rather than concentrated in the first few. This contrasts with the common implicit "spike" assumption behind low-rank adjustment: if confounding variance were concentrated in a few dominant components, removing them would collapse the background. Instead, removing the top K PCs eliminates only a modest portion of the dependence structure unless K is large, which is why residual correlations remain broadly distributed even at $K = 10$. Choosing a large K is not a remedy: PCs beyond the leading ones are estimated with increasing noise, so removing too many replaces bias with variance (visible for Precinct at $K = 50$). We treat the approximately power-law spectrum as an empirical regularity; mechanism is outside scope. Robustness checks against artifactual explanations are in SI Appendix G.

2.4 Bridge lemma: the crud scale and its decay

We formalize the link from the eigenvalue spectrum to the residual correlation distribution. Theorem 2 (SI Appendix D) shows that under a Haar-like delocalization condition on the eigenvectors of the covariance, the SD of the residual correlation $\rho_{ij}^{(K)}$ across uniformly random pairs is well approximated by a ratio of eigenvalue tail sums:

$$\sigma_K := \frac{\sqrt{\sum_{k>K} \lambda_k^2}}{\sum_{k>K} \lambda_k}. \tag{1}$$

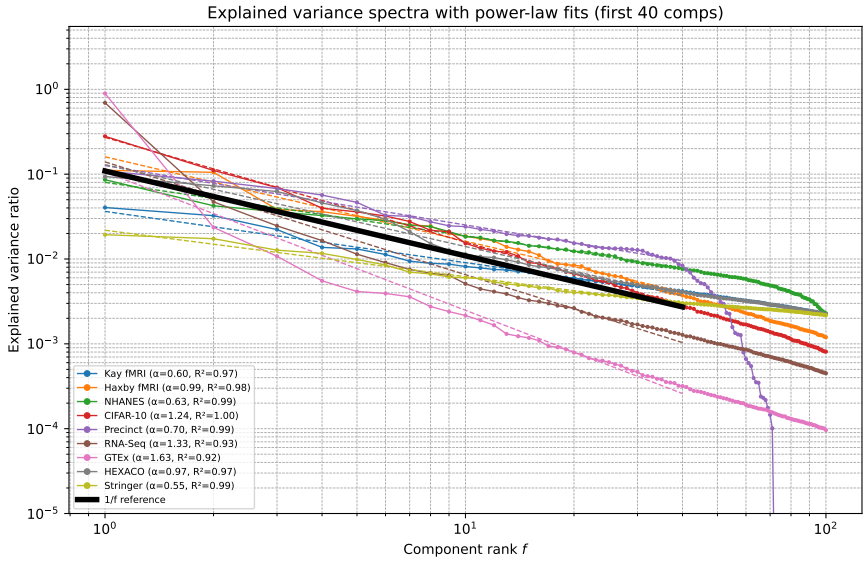

Figure 3: **Explained-variance spectra are approximately power-law.** Each point shows the variance a given PC explains (logarithmic axes). A straight line means a power law: variance decays as $k^{-\alpha}$. When α is near 1, variance spreads broadly across components, so removing the first few PCs leaves most of the shared dependence intact.

The numerator measures how unevenly the remaining variance is distributed across components; the denominator sets the scale. Eigenvector delocalization is met only approximately in our datasets (3.8–11.2× the Haar ideal; SI Appendix E, Test 2), but the resulting bias is conservative for the crud-scale claim (localization tends to make (1) overestimate the typical residual correlation; SI Appendix D.7).

If the eigenvalues follow a power law $\lambda_k \approx ck^{-\alpha}$, then σ_K decays slowly: in the especially relevant $\alpha \approx 1$ case, $\sigma_K \approx 1/(\sqrt{K}\log(p/K))$ (Corollary 1, SI Appendix D). Removing $K = 10$ components can leave a substantial residual crud scale when p is large, formalizing why low-rank adjustment shrinks the correlation distribution only modestly.

2.5 Decision-theoretic limit

A complementary question: how well can magnitude alone classify a pair as causally linked versus correlated by background dependence? Theorem 3 (SI Appendix D) gives the Bayes-error benchmark for this magnitude-only problem. Under a stylized Gaussian mixture in which a direct causal edge induces a mean shift μ in the adjusted association statistic T_{ij}, the minimum achievable misclassification error for any rule using only T_{ij} is $\Phi(-|\mu|/(2\sigma_K))$. When $|\mu| \lesssim \sigma_K$, this rate is a constant bounded away from zero. Two clarifications matter. First, this is the irreducibility of magnitude-only classification under this mixture, not a claim about what inference can recover about ρ_{ij} itself: an analyst with a sample correlation $\hat{\rho}_{ij}$ estimates ρ_{ij} ever more precisely as $n \to \infty$. The benchmark reflects overlap between the two classes in the mixture, not an obstruction to estimation. Second, whether the model-level σ_K matches the conceptually natural "background dependence" in a given domain depends on whether the chosen adjustment captures that domain's confounding structure (SI Appendix B). Sparse causal structure ($\pi \ll 1$) further compounds the magnitude-only limitation by collapsing positive predictive value (SI Appendix B.6). Heavy-tailed residuals, additional variance structure, and selected target pairs are discussed in SI Appendix D

and SI Appendix B.

2.6 Crud-aware testing

The empirical results above imply a minimum reporting standard for small-effect observational claims: compare any adjusted association to the same-pipeline empirical null—the distribution of adjusted associations over uniformly sampled feature pairs from the same dataset, computed under identical preprocessing and adjustment. The resulting crud-aware p-value is operationally a percentile rank: the fraction of same-pipeline reference pairs whose absolute adjusted association exceeds the target's. Despite the p-value notation, this is a magnitude-rank diagnostic rather than a frequentist test of a sharp null at $\rho = 0$; SI Appendix D clarifies the status. Associations below a chosen high percentile of this null (conventionally the 95th) should be reported as not distinctive against the domain background; the cutoff is conventional, not principled. The test is a magnitude diagnostic: failing to clear it means the magnitude alone cannot be cleanly distinguished from typical residual dependence, not that the relation is causally unimportant (mediation, suppression, and confounder-driven attenuation can produce small adjusted correlations from causally important relations); clearing it is necessary, not sufficient, for a causal interpretation. SI Section B gives details, the handling of selected target pairs (which violate exchangeability), stratified comparisons, and the relation to multiple-testing corrections.

2.7 A parametric crud-aware z-test

Many studies measure only a few variables on a small sample. Such a study can be understood as a small submatrix of the much larger data matrix one could have collected; the crud factor operates at the domain level. Under a Gaussian approximation (Theorem 1, SI Appendix D), the total variance of r for a random pair from the domain is the sum of crud and sampling contributions, $\mathrm{Var}(r) \approx \sigma_K^2 + 1/(n-1)$, where σ_K is the population residual-correlation SD defined in Eq. 1. This yields the test statistic

$$z_{\mathrm{crud}} = \frac{|r|}{\sqrt{\sigma_K^2 + 1/(n-1)}}, \tag{2}$$

with two-sided p-value $p_{\mathrm{crud}} = 2\,\Phi(-z_{\mathrm{crud}})$. The classical correlation test is the special case $\sigma_K = 0$. As n grows, $1/(n-1)$ shrinks but σ_K^2 does not, so for the purpose of distinguishing small associations from background dependence there are diminishing returns past $n \approx 1/\sigma_K^2$ (SI Appendix B.2). The Gaussian formula is broadly calibrated at the conventional 5% threshold and loses calibration for $p_{\mathrm{crud}} < 0.01$ in heavy-tailed domains; for stronger claims use the empirical null (SI Appendix D, Table D.1; SI Appendix B.3). A step-by-step reporting workflow, an online calculator, and guidance on choosing σ_K for an applied study are in SI Appendix B.4.

2.8 Worked example: folate–mortality in T2D

Liu et al. [2022] report a nominally significant association between folate and mortality in adults with type-2 diabetes ($r \approx 0.02$, $p_{\mathrm{classical}} \approx 0.04$, $n \approx 8{,}000$). With NHANES $\sigma_K = 0.093$ at $K = 10$, the crud-aware z-test gives $z_{\mathrm{crud}} = 0.24$, $p_{\mathrm{crud}} = 0.81$: the effect's magnitude is typical of randomly chosen NHANES pairs after the same adjustment. By contrast, within-facet HEXACO item pairs (designed to load on the same trait) clear the threshold easily ($p_{\mathrm{crud}} = 0.03$ for the median pair). SI Appendix B.9 gives both examples in full and reports two further cases (NHANES biomarkers, GTEx cis-eQTL effects).

3 Discussion

When reported adjusted associations are comparable to the crud scale, causal interpretation is fragile without additional design leverage, even if classical p-values are tiny. The crud-aware test provides a minimal magnitude standard for deciding when a small association is worth interpreting. As positive controls, smoking and lung cancer (RR$\approx$ 7; converted to a Pearson correlation via Eq. (16) of SI Appendix D.3 at representative population prevalences p_X=0.20, p_Y=0.06, this gives $r \approx 0.28$, or $\approx 3\sigma_K$ against NHANES at $K = 10$) and within-facet HEXACO personality items (best pair >99.99th percentile) clear the threshold; SI Appendix C.1 gives the structural-ceiling caveat for binary correlations and the case-control-vs-cohort distinction. Crucially, $3\sigma_K$ alone is not what makes smoking–cancer compelling: decades of converging dose–response, cessation, mechanism, and animal evidence carry that case. The crud-aware test is a magnitude diagnostic, not a sole filter for causal credibility.

Scope and leverage. Prior knowledge or design-based leverage effectively isolates variables from background dependence. Randomized treatments, discontinuities, policy changes, and valid instruments provide leverage that is orthogonal to generic correlation structure. Informed selection of which pairs to examine—from prior biology, mechanism, or pathway databases—narrows the empirical null; our impossibility result applies to the association statistic itself, not to the prior that selected the pair. Our results do not argue against such approaches; they target association-only workflows that interpret small adjusted pairwise correlations as causal evidence in the absence of additional leverage. The Bayes-error benchmark of Theorem 3 (SI Appendix D) bounds magnitude-only classification under a stylized mixture; it is not a claim about inference about ρ_{ij} itself, which improves with n, nor about systems that exploit multivariate structure, nonlinear dependence, mechanistic priors, or interventions. The homogeneity of broad spectra and slow σ_K shrinkage across our nine datasets is consistent with the obstacle generalizing beyond a single domain or modality; confirming truly cross-domain status requires a broader survey than nine cases can provide.

Empirical comparison to standard alternatives. On a synthetic data-generating process with latent-factor confounding and a small set of embedded direct-edge pairs (SI Appendix B.7), the crud-aware diagnostic at $K = 10$ recovers 96% of direct edges at the conventional 95th-percentile threshold, against 48 true positives and 947 false positives (30$\times$ better precision than classical $|r| > 1.96/\sqrt{n-1}$ and far better than BH-FDR, which fails to control its nominal FDR under structural confounding). The benchmark is a single stylized DGP rather than a method-dominance claim; in regimes with sparse rather than broadly-distributed confounding, BH-FDR will outperform. The takeaway is that the crud-aware diagnostic adds a useful magnitude check in the empirical regime the rest of this paper documents.

Relation to the causal-inference framework. The crud-aware test does not replace identification arguments. The standard causal-inference framework—directed acyclic graphs, the do-calculus, instruments, regression discontinuities, sensitivity analyses for unobserved confounding [Pearl, 2009, Imbens and Rubin, 2015, Cinelli and Hazlett, 2020, Oster, 2019]—is the primary apparatus for translating observational data into causal claims. A small adjusted correlation can be perfectly informative if the DAG identifies the effect; a large one can be uninformative if confounding is severe. Our test is a magnitude diagnostic complementary to identification: it asks "is the magnitude distinctive against same-pipeline residual dependence," which is a different question

from "is the effect identified." The test is most useful when identification is uncertain (the typical observational-claim setting), where it adds a magnitude floor on top of an unverified identification argument; it is least useful when identification is clean (e.g., a well-designed RCT), where magnitude alone is already meaningful relative to a sharp null.

Sensitivity of σ_K to discretionary choices. The reported quantitative multiples (r/σ_K, etc.) depend on K, preprocessing, design type, and reference-dataset choice. The qualitative conclusion—that the residual correlation distribution remains broadly distributed across these choices, and that small adjusted correlations rarely clear the conventional threshold against an approximately matched reference—is robust: σ_K at $K = 10$ and $K = 50$ differ by less than a factor of two in seven of nine datasets (Table 1), and the alt-adjustment results (SI Appendix G.4) confirm that Lasso does not collapse the residual distribution either. The quantitative pass/fail at the conventional threshold for a specific paper, however, is sensitive enough that we recommend reporting conclusions across a range of plausible σ_K values rather than a single number (SI Appendix B.4, SI Appendix B.8).

What to do when an effect does not clear the test. Failing the test is not a verdict that the relation is causally unimportant; it is a verdict that the magnitude alone cannot carry the case. Constructive next steps are: (i) seek design leverage—randomization, instruments, regression discontinuities, negative controls—which breaks the generic-dependence problem the test diagnoses; (ii) pre-register the hypothesis and replicate in an independent dataset, which is the most reliable correction for selection-driven inflation; (iii) accept that converging evidence (mechanism, dose–response, removal effect, prior literature) can carry a causal claim even when magnitude alone does not, as in the smoking–cancer case. When none of these are available and the magnitude is in the bulk of the empirical null, the right reporting standard is to label the association as not distinctive against background and treat any causal interpretation as provisional.

Limitations. All results concern linear dependence (correlations, linear PC adjustment, covariance spectra). In domains with substantial nonlinear dependence—gene regulatory networks, neural population codes, epidemiological dose–response—σ_K is a lower bound on the true background dependence; whether nonlinear adjustment would shrink the nonlinear background faster than linear adjustment shrinks the linear background is not tested here. The quantitative σ_K varies across domains and preprocessing; we use PC regression as a deliberately generic adjustment, not a claim of causal identification. In high-dimensional regimes (p not small relative to n), estimating dominant directions adds uncertainty to σ_K that the test does not propagate (SI Appendix G.4). The decision-theoretic limit is per-pair, equal-variance, Gaussian, and fixed-K. SI Appendix C.2 discusses asymmetric costs of skepticism in concrete settings (e.g., NIH R01 marginal-grant analysis). Three foundational issues are out of scope here and remain open: an analytic correction for selected target pairs (we recommend pre-registration and stratified references but provide no closed-form fix); whether the empirical null reflects latent dependence or unmodeled adjustment (we proceed under the practical position that magnitude typical of same-pipeline pairs is not by itself causal evidence, regardless of which interpretation prevails); and broader robustness of the impossibility theorem beyond the stylized Gaussian-mixture model.

Materials and Methods

Datasets. Nine large public datasets spanning population health, neuroscience, genomics, psychology, political science, and vision (Section 2.1; SI Appendix A gives provenance and dimensions): NHANES, precinct voting + Census, GTEx v8, Allen Institute mouse RNA-seq, HEXACO, CIFAR-10, Kay fMRI, Haxby fMRI, and Stringer mouse V1.

Preprocessing. Each feature is z-scored (mean zero, unit variance). Sample sizes used in analysis are capped at $n \approx 10{,}000$ per dataset for computational tractability where the raw n is larger.

Adjustment. The deliberately generic adjustment is regression on the top K principal component scores, with $K = 10$ as the headline choice and sensitivity reported across $K \in \{0, 1, 10, 50\}$ (Table 1). PCA is computed via randomized SVD with up to 300 components. Alternative adjustments (Lasso) are reported in SI Appendix G.4.

Empirical null and crud scale. For each dataset and each K, we compute residual correlations across uniformly sampled feature pairs (or, where $\binom{p}{2} \leq 5 \times 10^5$, all pairs). The crud scale σ_K is the SD of these residual correlations. Eigenvalue spectra are fit by power-law on the first 40 components against an exponential alternative via AIC.

Statistical analysis. The crud-aware z-test compares an observed correlation r to σ_K via Equation (2). The full empirical null replaces the Gaussian formula with the empirical fraction of reference pairs whose absolute adjusted association exceeds the target's. Calibration of the parametric formula against the empirical null in heavy-tailed datasets is reported in SI Appendix D, Table D.1. A reference catalog mapping common application domains to a recommended σ_K is in SI Appendix B.8.

Code and data. All analysis is in the Python package at `https://github.com/koerding/crud`. An interactive demonstration and a standalone crud-aware test calculator are at `https://koerding.github.io/crud/` and `https://koerding.github.io/crud/test.html`.

Code and Data Availability

Code and the interactive demo are available at `https://github.com/koerding/crud` and `https://koerding.github.io/crud/`. A standalone crud-aware test calculator is at `https://koerding.github.io/crud/test.html`. The draft PDF is at `https://raw.githubusercontent.com/koerding/crud/master/paper/crud.pdf`.

Supporting Information

A Datasets and provenance

The nine datasets listed in Section 2.1 were chosen to span dimensionalities, sample sizes, and measurement modalities so that any pattern common to all is unlikely to be an artifact of a single data type.

NHANES. U.S. National Health and Nutrition Examination Survey [National Center for Health Statistics, 2023]. Continuous, ordinal, and binary survey items: clinical biomarkers, demographics, health-behavior responses. $n \approx 30{,}000$, $p = 165$.

Precinct-level voting and Census demographics. 2016 U.S. precinct-level voting returns merged with Census American Community Survey demographics [Hill et al., 2019]. Continuous demographic and political variables. $n \approx 29{,}000$, $p = 83$.

GTEx v8 RNA-seq. Bulk gene-expression measurements from human skeletal muscle samples in the Genotype-Tissue Expression atlas [The GTEx Consortium, 2020]. Top $p = 803$ genes by variance, $n \approx 10{,}000$ samples after subsampling.

Allen Institute mouse brain RNA-seq. Single-cell expression from mouse cortex [Tasic et al., 2018]. $p = 387$ genes, $n \approx 10{,}000$ cells.

HEXACO. Personality-inventory item-level responses from a public dataset [Ashton and Lee, 2007]. $p = 242$ items, $n \approx 23{,}000$ respondents.

CIFAR-10. Small color natural images [Krizhevsky, 2009]. Pixel intensities, $p = 3{,}072$, $n = 30{,}000$.

Kay fMRI. Voxelwise BOLD responses to natural images in human visual cortex [Kay et al., 2008]. $p = 1{,}870$ voxels, $n \approx 8{,}400$ trials.

Haxby fMRI. Voxelwise BOLD responses to object categories in ventral temporal cortex [Haxby et al., 2001]. $p = 1{,}452$ voxels, $n \approx 24{,}000$ trials.

Stringer mouse V1. Two-photon imaging of mouse primary visual cortex [Stringer et al., 2019]. $p = 6{,}000$ neurons, $n \approx 7{,}000$ time points.

Convenience sampling. The nine were chosen for size, public availability, and modality coverage. They are not a probability sample of empirical sciences. Patterns common to all are evidence that the phenomenon is not data-modality-specific, but generalization to fields and datasets we did not analyze is an empirical question that nine cases cannot resolve.

Counterexamples we did not observe. A counterexample to the headline empirical finding would be a domain where a realistic adjustment used by practitioners collapses the residual correlation distribution, implying a rapidly decaying σ_K. Across the adjustment methods we tested—PC regression and Lasso (Appendix G.4)—neither collapsed the residual distribution on any of the nine datasets; whether some practitioner-favored adjustment in another domain would do so is not addressed by our experiments.

B Crud-aware testing: details and worked examples

This section gives the operational details of the crud-aware test, the parametric small-study version, the calibration of the Gaussian formula against the empirical null, the handling of selected target pairs and stratified comparisons, and the full worked examples summarized in the main text.

B.1 Empirical null in detail

For a target pair (i, j) with an adjusted association statistic (for example, a residual correlation after removing K PCs), we compare its absolute value to the empirical distribution of absolute adjusted associations over uniformly sampled feature pairs from the same dataset, computed under the identical preprocessing and adjustment pipeline. The resulting crud-aware p-value is the fraction of random pairs whose absolute adjusted association exceeds the target's; the crud percentile is its complement. The comparison is magnitude-based (two-sided): we rank by absolute value. For a uniformly random target pair processed through the same pipeline, this empirical p-value is (approximately) uniform by construction (Appendix D). For non-randomly chosen targets, the same test continues to control Type-I error *only under the null* that the pair's true association is exchangeable with the domain background. Practically, reporting an adjusted association should include its percentile within the empirical null; associations below a chosen high percentile of that null (e.g., the 95th) should be reported as not distinctive against the domain background, leaving open whether other lines of evidence—design leverage, mechanism, replication—support a causal interpretation. The 95th and 99th are conventional thresholds, not principled ones; the appropriate cutoff depends on prior odds and decision costs.

What the test does and does not claim. Concretely, the crud-aware p-value is a percentile rank: the fraction of same-pipeline reference pairs whose adjusted association magnitude exceeds the target's. It *does* answer "is this association magnitude distinctive within the residual structure of this dataset, given the chosen adjustment." It *does not* answer "is the underlying causal effect distinguishable from zero," "is the residual structure caused by latent confounding rather than incomplete adjustment," or "is this association causally informative for the substantive question." Those are separate questions that require design leverage, mechanistic models, or external validation. A target whose magnitude clears the threshold has earned only that its size cannot be explained as typical of the same-pipeline background—a necessary, not sufficient, condition for a causal interpretation.

Selected target pairs. Real applications test pairs chosen because of prior knowledge or pilot evidence. Selection violates the exchangeability assumption that justifies the empirical p-value as a Type-I rate: a pair flagged by a prior scan of the same data is not exchangeable with a uniformly random pair. Three practical recommendations: (i) when feasible, choose target pairs from a pre-registered hypothesis or from external evidence (mechanism, prior literature, an independent dataset), so the selection is independent of the reference data; (ii) when the target was selected from the same data, report the crud-aware p-value as a magnitude-rank statistic rather than a Type-I-controlled test, alongside the standard multiple-testing correction over the universe from which the pair was selected; (iii) where features fall into natural strata (modality, gene family, region), compute the empirical null over the within-stratum reference distribution so the comparison matches the target's structural class. None of these fully restores Type-I control under arbitrary selection; they reduce, not eliminate, the bias.

Distinction from multiple testing. Unlike multiple-testing corrections, which assume a sharp null of zero effect, the crud-aware test uses an empirical null built from same-pipeline reference pairs whose residual associations are typically nonzero. It controls Type-I error under the empirical null—the hypothesis that the target pair's association is typical of the background—rather than under the classical null of zero effect. The approach is related to Efron's empirical null framework [Efron, 2004], which estimates the null distribution of test statistics from data rather than assuming

the theoretical $\mathcal{N}(0,1)$. The key difference is that Efron's empirical null corrects for correlation-induced inflation of null z-scores, whereas the crud empirical null reflects residual dependence that survives generic adjustment—whether that dependence stems from genuine latent structure or from incomplete adjustment is left open by the test, which only asks whether the target association is distinctive relative to the same-pipeline background.

Stratified comparisons. When dependence is heterogeneous, sample random pairs within the relevant comparison class (same modality, ROI, gene family, measurement type) so the baseline matches the target claim, and report which stratum was used.

B.2 Diminishing returns beyond $n \approx 1/\sigma^2_{\text{crud}}$

The crud-aware framework yields a concrete insight about sample size. The z-test denominator $\sqrt{\sigma^2_{\text{crud}} + 1/(n-1)}$ reveals a natural crossover. When n is small ($n \ll 1/\sigma^2_{\text{crud}}$), sampling noise dominates; larger samples genuinely increase power to detect effects. But once n is large enough that the sampling noise $1/(n-1)$ is small relative to the crud variance σ^2_{crud}, the denominator is approximately σ_{crud} regardless of n: the bottleneck is no longer imprecise estimation but the fact that background correlations are themselves nonzero. Additional observations measure the crud more precisely but do not help separate signal from background. Because $\sigma_{\text{crud}} \approx 0.02$–$0.14$ across the domains in Table 1 (at $K = 10$), the crossover occurs at roughly $n \approx 50$–$2{,}500$. Beyond this point, classical p-values continue to shrink—explaining why massive studies routinely report "significant" but tiny effects—while the crud-aware p-value plateaus. For this specific purpose, there are sharply diminishing returns to collecting more than a few thousand observations beyond the crossover; this does not apply to the many other reasons a study might collect more data—improving precision on a known effect, supporting subgroup analysis, replicating a published finding, or detecting effects that are themselves much larger than the crud scale. Although larger n can in principle support larger K, in high-dimensional data K grows slowly relative to p and estimation noise in adjustment accumulates, so σ_K shrinks too slowly to close the gap.

B.3 Calibration of the parametric test

The Gaussian approximation underlying the parametric z-test is only approximate; in datasets where the off-diagonal residual correlation distribution is heavy-tailed, the parametric formula's bias is non-monotonic in $|r|$. We quantify this in Appendix D (Table D.1) for the two most kurtotic of our nine datasets (NHANES, RNASeq). The headline: at the conventional 5% threshold ($|r| \approx 2\sigma_{\text{crud}}$) the parametric formula is broadly calibrated; below $p_{\text{crud}} \approx 0.01$ it understates true tail mass by orders of magnitude. Read parametric p-values literally only at the conventional threshold; for stronger claims, use the full empirical null.

B.4 Step-by-step usage

How to use the crud-aware z-test.

1. Obtain the sample correlation r and sample size n from the study.
2. Look up a domain-appropriate crud scale σ_{crud}: use the residual correlation SD from a large reference dataset in your field (Table 1 reports values for various domains at $K = 10$), or consult published values.
3. Compute $z_{\text{crud}} = |r|/\sqrt{\sigma_{\text{crud}}^2 + 1/(n-1)}$ and compare to standard normal critical values ($z > 1.96$ for two-sided $p < 0.05$). Report the crud-aware p-value alongside the classical one.

We recommend reporting conclusions for a range of plausible σ_{crud} values as a sensitivity analysis. When a large reference dataset is available, the full empirical null is preferable, especially for $p_{\text{crud}} < 0.01$ where the parametric formula loses calibration in heavy-tailed domains (Appendix D, Table D.1).

Choosing the reference σ_{crud}. The test is sensitive to which reference is used, and the natural choice—the same dataset the target study used—is not always available. Three guidelines: (i) when the source data are accessible, compute σ_{crud} on the same dataset and, where the target study restricts to a sub-population (e.g., a T2D-only cohort within NHANES), compute it on the matching restriction, since sub-populations can have different residual dependence structure. (ii) When source data are not accessible, use a published σ_{crud} from a same-domain reference dataset of comparable design (Table 1), and report the conclusion across a range of plausible σ_{crud} values. (iii) Avoid mismatched designs: a case-control reference σ_{crud} for a population-cohort target (or vice versa) gives a misleading multiple, since case-control sampling inflates focal correlations involving the case-defining variable without proportionally inflating the random-pair σ_{crud} that anchors the test (Appendix D.3, caveat (vii)). An online calculator implementing this test is at `https://koerding.github.io/crud/test.html`.

B.5 Frequentist reading of the bridge lemma

The formal results in Appendix D can be read in a frequentist way by treating the spectrum and eigenvectors as fixed but unknown properties of the data-generating distribution; the distributional assumptions summarize empirical regularities across domains rather than introduce a Bayesian prior.

B.6 Base-rate compounding of the impossibility

The base rate of direct causal edges makes the impossibility of Theorem 3 worse in practice. In a gene-expression study measuring $p = 20{,}000$ genes, there are roughly 200 million possible gene pairs, but each gene directly regulates only a handful of others, so the number of direct regulatory edges is on the order of p, not p^2. More generally, each newly measured variable has a bounded number of direct causes and effects, so the prior probability π that a randomly chosen pair shares a direct causal link is $O(1/p)$—vanishingly small in high-dimensional datasets. Sparse causal structure ($\pi \ll 1$) compounds the spectral limitation by collapsing positive predictive value even at moderate signal-to-crud ratios: not only is each per-pair decision noisy, but even "significant" pairs are overwhelmingly likely to be noncausal.

B.7 Synthetic benchmark: comparison with multiple-testing alternatives

To check whether the crud-aware diagnostic adds anything over standard multiple-testing alternatives, we compare it against classical $|r|$ thresholding, Bonferroni, and Benjamini–Hochberg FDR [Benjamini and Hochberg, 1995] on data with a known causal-edge structure embedded in latent confounding.

Data-generating process. We simulate $p = 200$ features and $n = 2{,}000$ observations:

- $L = 10$ latent factors, each loading on a random sparse subset of 40 features with loadings $\sim \mathcal{N}(0, 0.40^2)$.
- $E = 50$ randomly placed direct-edge pairs; for each, an independent Gaussian source of SD 0.40 contributes additively to both endpoints.
- Independent unit-variance noise on each feature.

This produces a covariance matrix with broad latent-factor confounding among all variables (the empirical $\sigma_0 = 0.066$, comparable to several real datasets in Table 1) plus a small set of direct edges as the embedded "truth."

Methods.

1. Classical: $|r| > z_{0.025}/\sqrt{n-1}$.
2. Bonferroni: classical at α/M, $M = p(p-1)/2$.
3. BH-FDR at $q = 0.05$.
4. Crud-aware empirical at $K = 0$, 95th percentile of $|r|$.
5. Crud-aware empirical at $K = 10$, 95th percentile of $|r_{\text{resid}}|$.
6. Crud-aware empirical at $K = 10$, 99th percentile of $|r_{\text{resid}}|$.

Results. Table B.1 reports flagged count, true positives, false positives, precision, recall, and F1 for one run (seed fixed for reproducibility; reproducer in `scripts/synthetic_benchmark.py`).

Table B.1: Method comparison on the synthetic DGP. Truth: $E = 50$ direct edges out of $M = 19{,}900$ off-diagonal pairs (prior $P(\text{edge}) = 0.0025$). Empirical $\sigma_0 = 0.066$; $\sigma_{K=10} = 0.029$.

Method	Flagged	TP	FP	Precision	Recall	F1
Classical $\lvert r\rvert > 1.96/\sqrt{n-1}$	4,478	48	4,430	0.011	0.96	0.02
Bonferroni (α/M)	1,851	32	1,819	0.017	0.64	0.03
BH-FDR ($q = 0.05$)	3,277	48	3,229	0.015	0.96	0.03
Crud-aware, 95th pct, $K = 0$	995	9	986	0.009	0.18	0.02
Crud-aware, 95th pct, $K = 10$	995	48	947	0.048	0.96	0.09
Crud-aware, 99th pct, $K = 10$	199	40	159	0.20	0.80	0.32

Interpretation. Three findings.

(i) *Classical multiple-testing methods fail to control false discoveries when confounding is structural.* BH-FDR is supposed to control the false discovery rate at $q = 0.05$, but here it returns 3,229 false discoveries against 48 true ones (FDR ≈ 0.985, far above 0.05). The reason is well-known: BH assumes the p-values under the null are independent or weakly dependent, which fails badly under shared latent factors. Bonferroni is more conservative on flagged count but still recovers more false positives than true ones because the latent-factor correlations are large enough to clear even the Bonferroni threshold. Both methods are ill-suited to the regime the crud-aware test is designed for.

(ii) *Crud-aware testing without PC adjustment ($K = 0$) does not help.* At $K = 0$ the empirical null is dominated by latent-factor correlations of similar magnitude to direct-edge correlations, so the rank diagnostic flags many background pairs as "distinctive" and misses many direct edges. The PC adjustment is essential: removing the top $K = 10$ PCs shrinks σ_K from 0.066 to 0.029, and direct edges then dominate the upper tail of the residual correlation distribution.

(iii) *Crud-aware testing at $K = 10$ provides the best precision/recall tradeoff at conventional thresholds.* The 95th-percentile rule recovers 96% of direct edges with 4.8% precision (still low because the prior is 0.0025, but 30× better than classical). The 99th-percentile rule trades recall for precision and lifts F1 from 0.09 to 0.32.

Caveats. The DGP is stylized, with fixed loadings, Gaussian noise, and edge-strengths chosen to make the comparison readable. We do not claim the crud-aware test dominates alternatives in all regimes—if confounding is sparse and concentrated rather than broadly distributed, BH-FDR will perform much better. The takeaway is narrower: in regimes where the residual correlation distribution is broad (the regime the empirical sections of this paper document across nine real datasets), the crud-aware diagnostic provides a useful magnitude check that classical and FDR-based methods do not.

B.8 Reference catalog: choosing σ_K for an applied study

When the source data underlying a target study are not accessible, the user must select a σ_K from a published reference. Table B.2 maps common application domains to the closest reference dataset among the nine in this paper at $K = 10$, with caveats. The mapping is approximate: the right comparison would use a same-pipeline same-population σ_K, and the table is meant as a starting point, not a substitute for sensitivity analysis across plausible values.

A user whose application falls outside this list should pick the closest match by data modality and dimensional regime (p relative to n) and report sensitivity to a range of σ_K values that brackets neighboring entries. The catalog is a starting point; building a domain-matched empirical null on the same dataset is preferable whenever feasible.

B.9 Worked examples in full

The two examples below are illustrative endpoints—a psychometric within-construct pair (designed to load on the same latent trait) and a borderline-significant epidemiological finding—chosen to bracket the magnitude range, not as random draws from the literature.

A large effect that clears the crud scale. In the HEXACO personality inventory ($p = 242$ items, $n = 100{,}000$), items within the same personality facet are designed to measure the same underlying trait. The median within-facet correlation is $|r| = 0.32$ at $K = 0$, and 55% of within-facet pairs fall in the top 5% of the empirical null (11× enrichment over chance). Applying the

Table B.2: Approximate σ_K at $K = 10$ for common application domains, paired with the closest reference dataset among the nine in this paper. Caveats: this is a rough domain-similarity match, not a same-pipeline calibration; for any specific study the right σ_K is computed on the same data (or a matched sub-population) where possible. We recommend reporting conclusions across a range of plausible σ_K values, especially when the closest reference is itself a rough match.

Application domain	Closest reference	σ_K at $K = 10$
Observational epidemiology / health-survey	NHANES	0.093
Nutritional epidemiology / cancer registries	NHANES	0.093
Bulk gene expression / cis-eQTL (tissue)	GTEx	0.090
Single-cell or cell-type expression	Allen RNA-Seq	0.071
Brain-wide association studies (fMRI)	Haxby fMRI	0.077
Visual-cortex voxel responses	Kay fMRI	0.035
Single-neuron / population-coding	Stringer V1	0.021
Personality / individual differences	HEXACO	0.046
Natural images / pixel-level	CIFAR-10	0.105
Political / Census / area-level demographics	Precinct	0.136

crud-aware z-test with $\sigma_{\text{crud}} = 0.144$ (HEXACO at $K = 0$) gives $z_{\text{crud}} = 2.2$ for the median pair, $p_{\text{crud}} = 0.03$. After removing 10 PCs, 29% of within-facet pairs remain in the top 5% (6× enrichment). Known-strong associations clear the empirical null.

A small effect that does not. In Liu et al. [2022], multivariable models report a nominally significant association between folate and mortality in adults with T2D (folate quartile 1 vs 2 HR 1.17, 95% CI 1.01–1.37) in an NHANES-derived cohort of adults with type 2 diabetes ($n \approx 8{,}000$). Converting from the reported confidence interval via $z = \log(\text{HR})/\text{SE}(\log \text{HR})$ and $r \approx z/\sqrt{n}$ gives $r \approx 0.02$, giving $p_{\text{classical}} \approx 0.04$. Applying the crud-aware z-test with $\sigma_{\text{crud}} = 0.093$ (NHANES at $K = 10$) gives $z_{\text{crud}} = 0.24$, $p_{\text{crud}} = 0.81$. Even this barely significant classical result becomes a crud-aware p-value of 0.81: the effect sits squarely in the middle of the empirical null—its magnitude is typical of randomly chosen NHANES variable pairs after the same adjustment. For comparison, known strong biomarker pairs in the same dataset (hemoglobin vs hematocrit, $r = 0.91$; ALT vs AST, $r = 0.83$) easily clear the crud scale.

Additional worked examples are in Appendix G.5: NHANES biomarkers (positive—known mechanistic pairs clear the crud scale easily) and GTEx RNA-seq (negative—most reported cis-eQTL effects do not).

C Discussion: extended caveats and stakes

C.1 Smoking–lung-cancer: structural-ceiling derivation and case-control vs cohort

Smoking and lung cancer are often cited as the canonical example of a strong, well-established epidemiological causal link. Pooled multiple-adjusted relative risks are about 7 in large cohort meta-analyses [O'Keeffe et al., 2018]. Converting via Equation (16) at representative population prevalences ($p_X = 0.20$ for smoking prevalence, $p_Y = 0.06$ for lifetime lung-cancer incidence) gives $r \approx 0.28$, roughly $3\,\sigma_K$ against the NHANES residual scale at $K = 10$ ($\sigma_K = 0.093$): the smoking–lung-cancer association clears the conventional crud-aware threshold but is, in correlation units,

much smaller than the relative-risk scale alone would suggest.

This is in fact close to the structural ceiling on binary correlations at this outcome prevalence:

$$\phi_{\max} = \sqrt{\frac{p_Y(1-p_X)}{p_X(1-p_Y)}} \approx 0.51 \quad \text{here, regardless of RR.} \tag{3}$$

Even an arbitrarily large causal effect on a rare binary outcome cannot produce a much larger population correlation in a cohort design. So smoking-cancer at $r \approx 0.28$ is over half the structural maximum, and the test correctly identifies it as exceptional given that ceiling.

Case-control summaries of the same data routinely report $r \approx 0.7$. That figure is a property of the 1:1 case-control sampling design (which artificially sets $p_Y = 0.5$), not of the population correlation, and is not the right comparison to a population-cohort σ_K such as NHANES (Appendix D.3, caveat (vii)). The σ_K-multiple is design-internal: case-control sampling inflates focal correlations involving the case-defining variable but leaves correlations among other variable pairs (which dominate σ_K) roughly unchanged, so the same underlying causal relationship can read as $\sim 3\sigma_K$ in cohort units and $\sim 7\sigma_K$ in case-control units.

A second example from a different domain illustrates the upper end without any binary-correlation ceiling: in the HEXACO literature, Dark Triad traits are strongly negatively correlated with Honesty-Humility (rs $= -0.72, -0.57, -0.53$); even the smallest of these, $|r| = 0.53$, is roughly $3.7\,\sigma_K$ at $K = 0$ in HEXACO ($\sigma_0 = 0.144$; Table 1) [Lee and Ashton, 2005]. These two textbook examples are illustrative endpoints, chosen to anchor the high end of the magnitude scale, not representative of the typical published effect.

C.2 Practical stakes: asymmetric costs of skepticism

The cost of treating small-effect causal claims skeptically depends on the downstream decisions they inform. Several of those decisions are themselves of modest causal consequence: a regression-discontinuity analysis of NIH R01 funding estimates that receipt of a marginal grant causes fewer than one additional publication over the following five years, with citation effects not statistically distinguishable from zero [Jacob and Lefgren, 2011]. The implication is an asymmetry—raising the bar for small-effect causal claims is unlikely to forgo large substantive gains, because the decisions such claims inform often have only small effects on the outcomes they are meant to improve.

D Bridge lemmas: crud scale and limits of association-only causal inference

The main text documents an empirical pattern—broad background correlations that persist after low-rank adjustment—and proposes testing associations against an empirical null derived from this background. Here we formalize why the pattern arises and what it implies. We prove two results. First, the typical residual correlation across random feature pairs after removing K principal components is controlled by a simple ratio of eigenvalue tail sums (Theorem 2). When eigenvalues decay as a power law, this ratio shrinks only as $\sim 1/\sqrt{K}$, explaining the slow empirical decay in Table 1. Second, we show that any decision rule based solely on a one-dimensional adjusted association statistic incurs irreducible classification error when the causal signal is comparable to the crud scale (Theorem 3). Both proofs rely on standard random-matrix tools—Haar-like eigenvector moment bounds and Gaussian signal detection—and yield closed-form expressions that can be checked against the empirical values. We take the approximately $1/f$ spectrum as an empirical

input; explaining why it arises mechanistically in any particular domain is outside the scope of this paper.

D.1 Setup and notation

Let $X \in \mathbb{R}^p$ be a mean-zero random vector with population covariance $\Sigma = \mathbb{E}[XX^\top] \in \mathbb{R}^{p\times p}$. Let $\Sigma = V\Lambda V^\top$ be an eigendecomposition with $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$ and $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. For an integer $K \in \{0, 1, \ldots, p-1\}$, let $P_K := \sum_{k=1}^{K} v_k v_k^\top$ be the projector onto the top-K eigenspace (with $P_0 = 0$). Define the population residual covariance after removing the top K principal components as

$$\Sigma^{(K)} := (I - P_K)\Sigma(I - P_K) = \sum_{k>K} \lambda_k v_k v_k^\top. \tag{4}$$

Define the population residual correlation for $i \neq j$ by

$$\rho_{ij}^{(K)} := \frac{\Sigma_{ij}^{(K)}}{\sqrt{\Sigma_{ii}^{(K)}\Sigma_{jj}^{(K)}}}. \tag{5}$$

D.2 Crud-aware p-values via the empirical null

A natural question is whether an observed adjusted association is unusual for the domain or merely typical background dependence. This subsection defines a "crud-aware" p-value that tests an observed adjusted association against the domain's empirical null. The key idea is to treat the realized dataset as fixed and define a reference distribution by drawing feature pairs uniformly at random, while running the same preprocessing and adjustment pipeline as for the target pair. The resulting empirical null provides an assumption-light baseline for whether an observed association is distinctive.

Adjusted association statistic. Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ denote the observed data matrix (columns are features). Fix an adjustment procedure A_K (for example: regress each feature on the top K principal component scores and standardize the residuals). For any feature pair (a, b) with $1 \leq a < b \leq p$, define a one-dimensional adjusted association statistic

$$T_{ab} = T(\mathbf{X}; a, b, K). \tag{6}$$

A canonical example is the sample correlation of adjusted (residualized) features, $T_{ab} = \widehat{\rho}_{ab}^{(K)}$. The only requirement is that the same pipeline (preprocessing, adjustment, and statistic) is used for the target pair and for reference pairs.

Crud-null reference distribution. Rather than testing whether T_{ij} is zero, we compare its magnitude to what is typical for a uniformly random feature pair under the same pipeline. Let

$$(A, B) \sim \mathrm{Unif}\Big(\{(a, b) : 1 \leq a < b \leq p\}\Big), \tag{7}$$

independent of all other randomness. For a threshold $t \geq 0$, define the (two-sided) crud-null tail probability

$$p_{\mathrm{crud}}(t) \;:=\; \mathbb{P}\big(|T_{AB}| \geq t \,\big|\, \mathbf{X}\big). \tag{8}$$

This is a conditional probability given the realized dataset $\mathbf{X}$, where the only randomness is the selection of a uniformly random feature pair.

Crud-aware p-value for a target pair. For a specific target pair (i, j), set $t_{\text{obs}} = |T_{ij}|$ and define

$$p_{\text{crud}}(i, j) \;:=\; \mathbb{P}\big(|T_{AB}| \geq |T_{ij}| \,\big|\, \mathbf{X}\big). \tag{9}$$

Equivalently, $p_{\text{crud}}(i, j)$ is the fraction of feature pairs whose absolute adjusted association equals or exceeds that of the target pair—a two-sided, magnitude-based rank. The *crud percentile* is the complement $1 - p_{\text{crud}}(i, j)$: values near 1 indicate that the target pair's association is unusually large relative to the domain background.

Status of p_{crud}. Despite the notation, p_{crud} is operationally a percentile-rank statistic of the target's magnitude within the same-pipeline reference distribution, not a frequentist p-value in the strict sense. The reference distribution is empirical and tied to the realized dataset $\mathbf{X}$; "rejecting" at the conventional 5% level means the target's magnitude exceeds the 95th percentile of that empirical reference, not that a sharp null at $\rho_{ij} = 0$ has been falsified. Under exchangeability between the target and the reference, p_{crud} is uniformly distributed by construction—this is a tautology of the rank operation, not a substantive Type-I guarantee. We retain the p-value notation because it is operationally familiar and the percentile reading is the practical content; readers who want a clean inferential reading should treat p_{crud} as a magnitude-rank diagnostic, not as a probability under a causal null.

Monte Carlo estimation. In practice, $p_{\text{crud}}(i, j)$ is approximated by sampling M reference pairs uniformly from all $\binom{p}{2}$ pairs, computing $T_{A_m B_m}$ for each, and taking the fraction (plus one, for the standard add-one correction) whose absolute value exceeds $|T_{ij}|$. For a uniformly random target pair, p_{crud} is (super-)uniform by a standard exchangeability argument. For a target pair selected non-randomly, the rank statistic remains uniformly distributed only under exchangeability with the reference. Selection by prior hypothesis or pilot data violates exchangeability—e.g., pairs flagged by an earlier scan of the same data are no longer exchangeable with random pairs—so for such targets p_{crud} remains a percentile diagnostic but loses any uniform-distribution interpretation, and does not by itself adjust for selection.

Practical refinements. When features are heterogeneous in type or scale, reference pairs should be stratified (e.g., within the same modality, ROI, or gene family) so the baseline matches the target claim. When crud-aware p-values are computed for many pre-specified pairs, standard multiplicity corrections (e.g., Benjamini–Hochberg) can be applied. The definition of p_{crud} depends only on the choice of adjustment pipeline and statistic; alternative adjustment strategies are compared in Appendix G.4.

A parametric crud-aware test for small studies. The empirical null above requires a data matrix large enough to estimate the background distribution of pairwise associations. Many studies, however, measure only a handful of variables on a small sample—a psychologist recruiting $n = 20$ participants and computing a single correlation, for instance. Such a study can be understood as observing a small submatrix of the much larger data matrix one *could* have collected had the budget permitted measuring all variables in the domain. The crud factor operates at the level of the full domain: if the typical background correlation among personality measures is $\sigma_{\text{crud}} \approx 0.15$, that background does not disappear simply because the investigator chose to measure only two of them. This motivates a parametric analogue of the crud-aware p-value that requires only summary statistics.

Setup. Suppose an investigator observes the sample correlation $\hat{\rho}$ between two variables measured on n independent observations. These two variables are drawn from a domain of p variables with population covariance Σ. Define the population correlation for the pair (i, j) as $\rho_{ij} = \Sigma_{ij}/\sqrt{\Sigma_{ii}\Sigma_{jj}}$, and let σ^2_{crud} denote the variance of ρ_{ij} when (i, j) is drawn uniformly at random from all $\binom{p}{2}$ pairs. The quantity σ_{crud} characterizes the spread of background correlations in the domain and can be estimated from large published datasets or reference analyses. (When K principal components have been removed, one uses the residual correlations $\rho^{(K)}_{ij}$ and the corresponding $\sigma^{(K)}_{\text{crud}}$.)

Theorem 1 (Parametric crud-aware test)**.** *Suppose the pair (I, J) is drawn uniformly at random from $\{(a, b) : 1 \le a < b \le p\}$, and conditionally on (I, J), the sample correlation $\hat{\rho}$ satisfies*

$$\hat{\rho} = \rho_{IJ} + \varepsilon, \qquad \varepsilon \mathrel{\dot{\sim}} \mathcal{N}(0, \sigma^2_n), \qquad \sigma^2_n \approx \frac{(1 - \rho^2_{IJ})^2}{n - 1} \approx \frac{1}{n - 1}, \tag{10}$$

where ε is independent of (I, J) and the approximation $\sigma^2_n \approx 1/(n - 1)$ holds when $|\rho_{IJ}|$ is small. Further suppose that the population correlation ρ_{IJ}, viewed as a random variable over the uniform draw of pairs, satisfies

$$\rho_{IJ} \sim \mathcal{N}(0, \sigma^2_{\text{crud}}). \tag{11}$$

Then the marginal distribution of the sample correlation under the crud null is approximately

$$\hat{\rho} \mathrel{\dot{\sim}} \mathcal{N}\left(0,\ \sigma^2_{\text{crud}} + \frac{1}{n - 1}\right). \tag{12}$$

Proof. Write $\hat{\rho} = \rho_{IJ} + \varepsilon$. By assumption, $\rho_{IJ} \sim \mathcal{N}(0, \sigma^2_{\text{crud}})$ and $\varepsilon \mathrel{\dot{\sim}} \mathcal{N}(0, 1/(n - 1))$. Under the approximation in (10) with $\sigma^2_n \approx 1/(n-1)$, the conditional variance of the sampling noise does not depend on ρ_{IJ}, allowing us to treat them as independent. The sum of two independent Gaussians is Gaussian, and their variances add:

$$\text{Var}(\hat{\rho}) \approx \text{Var}(\rho_{IJ}) + \text{Var}(\varepsilon) \approx \sigma^2_{\text{crud}} + \frac{1}{n - 1}. \tag{13}$$

Both components have zero mean, so $\mathbb{E}[\hat{\rho}] = 0$, yielding the approximate distribution in (12). □

The crud-aware z-test. Theorem 1 yields a direct test statistic. For an observed correlation r with sample size n and domain crud level σ_{crud}, define

$$z_{\text{crud}} = \frac{|r|}{\sqrt{\sigma^2_{\text{crud}} + \frac{1}{n-1}}}, \tag{14}$$

with two-sided p-value $p_{\text{crud}} = 2\,\Phi(-z_{\text{crud}})$. For comparison, the classical correlation test uses $z_{\text{classical}} = |r|\sqrt{n - 1}$, which is the special case $\sigma_{\text{crud}} = 0$: it tests against the null of zero population correlation, ignoring that a nonzero correlation may be entirely typical of the domain background.

When the correction matters. The ratio of the two test statistics is

$$\frac{z_{\text{crud}}}{z_{\text{classical}}} = \frac{1}{\sqrt{1 + (n - 1)\,\sigma^2_{\text{crud}}}}. \tag{15}$$

When n is small relative to $1/\sigma^2_{\text{crud}}$, this ratio is close to 1 and the crud correction is modest—though it still raises the significance threshold, reflecting that a correlation of $r = 0.3$ in a field with

$\sigma_{\text{crud}} = 0.15$ is less remarkable than the same r in a field with negligible background. When n is large, the sampling variance $1/(n-1)$ shrinks but the crud variance does not. A study with $n = 200$ and $r = 0.15$ yields $p < 0.05$ classically, yet $p_{\text{crud}} \approx 0.37$ when $\sigma_{\text{crud}} = 0.15$: the observed association is entirely typical of the domain background. In this regime, the classical test is detecting crud, not signal.

Assumptions and practical guidance. The derivation relies on two approximations. First, the Gaussian model (11) for the distribution of population correlations over random pairs. The audit (Appendix E, Test 5) finds that residual correlations are typically heavier-tailed than Gaussian, with excess kurtosis up to ≈ 35 in the most affected datasets (NHANES) and skewness up to ≈ 3. The bias direction is non-monotonic in $|r|$. Near the bulk the parametric formula *overstates* tail mass (because Gaussian tails fall off slower than the empirical tails right above zero, in datasets whose mass is concentrated near the center); in the deep tail the parametric formula *understates* tail mass (because the empirical distributions have heavy tails). To make this concrete, Table D.1 compares empirical and parametric two-sided crud-aware p-values at representative target $|r|$ in the two most kurtotic of our nine datasets:

Table D.1: Empirical vs. parametric two-sided crud-aware p-values at representative target $|r|$, in the two most kurtotic of our nine datasets at $K = 10$. Empirical p is the fraction of off-diagonal residual correlations with $|r_{\text{resid}}| \geq t$; parametric p uses Equation (14) with the corresponding σ_K and n.

	NHANES ($\sigma_K = 0.094$, kurt ≈ 37)			RNASeq ($\sigma_K = 0.071$, kurt ≈ 13)		
$\lvert r\rvert$	param. p	emp. p	ratio	param. p	emp. p	ratio
0.10	0.290	0.130	0.45	0.164	0.099	0.60
0.15	0.112	0.063	0.56	0.037	0.036	0.97
0.20	0.034	0.035	1.01	0.005	0.018	3.3
0.25	0.008	0.023	2.8	0.0005	0.011	22
0.30	0.0015	0.012	8.4	0.00003	0.008	270

The agreement is within a factor of ~ 1 (~ 3 in RNASeq) near the conventional 95th-percentile region $|r| \approx 2\sigma_K$. Deeper into the tail ($|r| \gtrsim 3\sigma_K$), the parametric formula understates true tail mass by one to several orders of magnitude. Practical implication: at the conventional 5% threshold the parametric test is broadly calibrated; below $p \approx 0.01$ its p-values should not be read literally—for those, use the full empirical null (Section D.2). We have no general analytic correction. Second, the linearization $\sigma_n^2 \approx 1/(n-1)$, which is standard for $|\rho|$ not too close to ± 1 and n not too small.

In practice, σ_{crud} must be estimated from external data, introducing additional uncertainty. We recommend reporting the conclusion for a range of plausible σ_{crud} values as a sensitivity analysis. The parametric version is intended for the common setting where only summary statistics are at hand—a simple corrective to the standard practice of testing against the independence null alone, with the heavy-tail caveat above.

D.3 Comparing scales: r, RR, and σ_K

Reported effect sizes in the empirical sciences arrive on multiple incommensurate scales: Pearson r, partial correlations, r^2, relative risk RR, odds ratio OR, hazard ratio HR, Cohen's d. To compare any of these to a crud scale σ_K (in correlation units), conversion is needed.

For a binary exposure $X \in \{0,1\}$ with prevalence $p_X = \Pr[X = 1]$ and a binary outcome $Y \in \{0,1\}$ with prevalence $p_Y = \Pr[Y = 1]$, the population Pearson correlation between X and Y corresponding to a relative risk $\mathrm{RR} = \Pr[Y = 1 \mid X = 1]/\Pr[Y = 1 \mid X = 0]$ is

$$r \;=\; \frac{p_X(1-p_X)\,(\mathrm{RR}-1)\,p_Y}{(\mathrm{RR}\,p_X + 1 - p_X)\,\sqrt{p_X(1-p_X)\,p_Y(1-p_Y)}}, \tag{16}$$

obtained by computing $\Pr[X = 1, Y = 1]$ from RR, p_X, p_Y and substituting into $r = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}$.

Equation (16) is sensitive to both prevalences. Table D.2 reports a single benchmark slice: $p_X = 0.20$, $p_Y = 0.06$, chosen as a rough match to a smoking-prevalence/lung-cancer-incidence-style setting, with r/σ_K taken against NHANES $\sigma_K(K = 10) = 0.093$ (the only one of our nine datasets that resembles a general-health observational cohort).

Table D.2: Conversion from relative risk to Pearson r at fixed prevalences ($p_X = 0.20$, $p_Y = 0.06$), and to multiples of the NHANES residual scale at $K = 10$ ($\sigma_K = 0.093$).

RR	r from (16)	r/σ_K
1.05	0.005	0.05
1.10	0.011	0.11
1.50	0.046	0.49
2	0.084	0.90
3	0.144	1.55
5	0.224	2.41
7	0.276	2.97
10	0.325	3.49

Caveats. This bridge is approximate and should be read as order-of-magnitude. (i) The mapping in (16) is exact only for binary X and Y. Many published RRs are reported from continuous, ordinal, or time-to-event exposures, where binary-RR-to-r is a heuristic. (ii) The output is sensitive to both p_X and p_Y. Equivalently, the conversion requires a baseline rate $p_0 = \Pr[Y = 1 \mid X = 0] = p_Y/(\mathrm{RR}\,p_X + 1 - p_X)$: choosing (p_X, p_Y) implicitly fixes p_0, and many published papers report RR with p_X but not p_Y, leaving the baseline rate to be supplied externally. Doubling p_Y at fixed RR can change r by a factor of $\sqrt{2}$ or more; halving p_X from 0.5 to 0.05 at fixed RR= 1.5 changes r from 0.10 to 0.04. Table D.2 therefore fixes one prevalence pair (with implied $p_0 \approx 0.027$ at RR= 7); reading other rows in the empirical literature requires recomputing with that domain's prevalences. (iii) The reference σ_K in the table is NHANES, which is observational health-survey data and not a smoking-cancer-specific reference; matched references would change r/σ_K. (iv) For odds ratios under low outcome prevalence, OR $\approx$ RR and (16) applies directly; for higher prevalences OR and RR diverge. (v) Hazard ratios from time-to-event analyses can be roughly converted via $r \approx \log(\mathrm{HR})/(\mathrm{SE}(\log \mathrm{HR})\sqrt{n})$ [as we use for the folate worked example, Liu et al., 2022, §2.8], but this depends on duration and censoring structure. (vi) For continuous standardized variables, r equals the standardized regression coefficient β; for r^2 to $|r|$ the conversion is $|r| = \sqrt{r^2}$. (vii) Case-control designs artificially set $p_Y = 0.5$ by 1:1 sampling of cases and controls, which inflates the Pearson correlation between exposure and outcome relative to the population value at the same RR; e.g., the smoking–lung-cancer correlation is often cited as $r \approx 0.7$ from case-control summaries, while the same RR in a population cohort with $p_Y \approx 0.06$ gives $r \approx 0.28$. The right comparison to σ_K uses a same-design reference, since σ_K inherits the design of the dataset it is computed

from. Note that σ_K-multiples are therefore design-internal and not invariant across designs: the same underlying causal relationship can give a $3\,\sigma_K$ multiple in cohort units and a $\sim 7\,\sigma_K$ multiple in case-control units, because case-control sampling inflates focal correlations involving the case-defining variable but leaves correlations among other variable pairs (which dominate σ_K) roughly unchanged. None of our nine reference datasets are case-control, so the population formula in (16) is the right bridge for our σ_K values.

We use (16) in two places: the smoking–lung-cancer comparison in the Discussion and the cross-field summary in Appendix F. Both treat the converted values as approximate.

D.4 Eigenvector delocalization assumption

To connect the eigenvalue spectrum to typical pairwise correlations, we need that eigenvectors are not concentrated on small sets of coordinates.

Assumption 1 (Haar-like eigenvectors / delocalization)**.** *The eigenvector matrix V is distributed as a Haar-uniform random orthogonal matrix, independent of Λ, or (more generally) satisfies the moment bounds of Haar orthogonal matrices: for all $i \neq j$ and all k,*

$$\mathbb{E}[v_{ik}] = 0, \qquad \mathbb{E}[v_{ik}^2] = \frac{1}{p}, \qquad \mathbb{E}[v_{ik}^2 v_{jk}^2] = \frac{1}{p(p+2)}. \tag{17}$$

Additionally, cross-component correlations are negligible in the sense that the covariance of $v_{ik}v_{jk}$ and $v_{i\ell}v_{j\ell}$ for $k \neq \ell$ contributes only lower-order terms ($O(p^{-3})$) to the variance calculations below.

Assumption 1 is standard in random matrix theory as a proxy for "incoherent" or "delocalized" eigenvectors. Empirically, it is supported when no small subset of variables dominates principal components.

Remark (localized crud). If eigenvectors are partially localized or block-structured, the same calculations imply that crud may be localized rather than global: typical residual correlations can remain large within blocks even if global random-pair correlations are smaller.

D.5 Theorem A.1: crud scale after removing top-K principal components

Theorem 2 (Crud scale under top-K PC removal)**.** *Assume Assumption 1. Fix $K \in \{0, \ldots, p-1\}$ and define tail sums*

$$T_1(K) := \sum_{k>K} \lambda_k, \qquad T_2(K) := \sum_{k>K} \lambda_k^2. \tag{18}$$

Then for a uniformly random pair (i, j) with $i \neq j$,

$$\mathrm{SD}\Big(\rho_{ij}^{(K)}\Big) = \frac{\sqrt{T_2(K)}}{T_1(K)} \cdot \big(1 + o_p(1)\big), \tag{19}$$

as $p \to \infty$ (with K allowed to grow with p, provided $K < p$ and the tail is nontrivial). Equivalently, the typical residual correlation magnitude across random feature pairs is controlled by

$$\sigma_K := \frac{\sqrt{\sum_{k>K} \lambda_k^2}}{\sum_{k>K} \lambda_k}. \tag{20}$$

Proof. Write the off-diagonal residual covariance entry as

$$\Sigma_{ij}^{(K)} = \sum_{k>K} \lambda_k v_{ik} v_{jk}, \qquad i \neq j. \tag{21}$$

By symmetry under Assumption 1, $\mathbb{E}[v_{ik}v_{jk}] = 0$, hence $\mathbb{E}[\Sigma_{ij}^{(K)}] = 0$.

For the variance, expand

$$\mathrm{Var}(\Sigma_{ij}^{(K)}) = \sum_{k>K} \lambda_k^2 \,\mathrm{Var}(v_{ik}v_{jk}) + 2 \sum_{K<k<\ell} \lambda_k \lambda_\ell \,\mathrm{Cov}(v_{ik}v_{jk}, v_{i\ell}v_{j\ell}). \tag{22}$$

Under the Haar moment bounds,

$$\mathrm{Var}(v_{ik}v_{jk}) = \mathbb{E}[v_{ik}^2 v_{jk}^2] - \mathbb{E}[v_{ik}v_{jk}]^2 = \frac{1}{p(p+2)}. \tag{23}$$

The cross-component covariance term is of smaller order (Assumption 1), so

$$\mathrm{Var}(\Sigma_{ij}^{(K)}) = \frac{1}{p(p+2)} \sum_{k>K} \lambda_k^2 \cdot \big(1 + o(1)\big), \tag{24}$$

and therefore

$$\mathrm{SD}(\Sigma_{ij}^{(K)}) = \frac{1}{p}\sqrt{T_2(K)} \cdot \big(1 + o(1)\big). \tag{25}$$

Next, consider the diagonal:

$$\Sigma_{ii}^{(K)} = \sum_{k>K} \lambda_k v_{ik}^2. \tag{26}$$

Taking expectations gives

$$\mathbb{E}[\Sigma_{ii}^{(K)}] = \sum_{k>K} \lambda_k \mathbb{E}[v_{ik}^2] = \frac{1}{p} T_1(K). \tag{27}$$

Under Assumption 1, $\mathrm{Var}(\Sigma_{ii}^{(K)}) \approx \frac{2}{p^2} T_2(K)$, so the relative fluctuation is $\mathrm{Var}(\Sigma_{ii}^{(K)})/\mathbb{E}[\Sigma_{ii}^{(K)}]^2 \approx 2\sigma_K^2$. Provided the eigenvalue tail is broad enough that $\sigma_K \to 0$ (which holds when $\alpha \leq 1$, or as $K \to \infty$ for any $\alpha > 1/2$), $\Sigma_{ii}^{(K)}$ concentrates around its mean with relative fluctuations $o_p(1)$. For fixed small K with $\alpha > 1$, σ_K remains a moderate constant and the formula below becomes an approximation whose quality is verified empirically in Appendix E (Test 4). Hence

$$\sqrt{\Sigma_{ii}^{(K)} \Sigma_{jj}^{(K)}} = \frac{1}{p} T_1(K) \cdot \big(1 + o_p(1)\big). \tag{28}$$

Finally, for $i \neq j$,

$$\rho_{ij}^{(K)} = \frac{\Sigma_{ij}^{(K)}}{\sqrt{\Sigma_{ii}^{(K)} \Sigma_{jj}^{(K)}}} = \frac{\Sigma_{ij}^{(K)}}{\frac{1}{p} T_1(K)} \cdot \big(1 + o_p(1)\big). \tag{29}$$

Taking standard deviations and substituting the expression for $\mathrm{SD}(\Sigma_{ij}^{(K)})$ yields

$$\mathrm{SD}(\rho_{ij}^{(K)}) = \frac{\frac{1}{p}\sqrt{T_2(K)}}{\frac{1}{p} T_1(K)} \cdot \big(1 + o_p(1)\big) = \frac{\sqrt{T_2(K)}}{T_1(K)} \cdot \big(1 + o_p(1)\big). \tag{30}$$

□

Corollary 1 (Power-law eigenvalues imply slow decay in K)**.** *Assume $\lambda_k = ck^{-\alpha}$ for $k = 1, \ldots, p$ with $c > 0$ and $\alpha > 1/2$, and assume the conditions of Theorem 2. Then for $1 \ll K \ll p$, the crud scale $\sigma_K = \sqrt{T_2(K)}/T_1(K)$ satisfies:*

- *If $\alpha = 1$, then $\sigma_K \asymp \dfrac{1}{\sqrt{K}\,\log(p/K)}$.*
- *If $\alpha > 1$, then $\sigma_K \asymp \dfrac{1}{\sqrt{K}}$.*
- *If $1/2 < \alpha < 1$, then $\sigma_K \asymp \dfrac{1}{\sqrt{K}} \left(\dfrac{K}{p}\right)^{1-\alpha}$.*

Here $a_K \asymp b_K$ means a_K/b_K is bounded above and below by positive constants depending only on (c, α).

Proof. By Theorem 2, it suffices to compute asymptotics of $T_1(K)$ and $T_2(K)$. Approximate sums by integrals.

If $\alpha = 1$: $T_1(K) = c\sum_{k=K+1}^{p} k^{-1} \asymp c\log(p/K)$ and $T_2(K) = c^2 \sum_{k=K+1}^{\infty} k^{-2} \asymp c^2/K$. Thus $\sigma_K \asymp (c/\sqrt{K})/(c\log(p/K))$.

If $\alpha > 1$: $T_1(K) \asymp c\int_K^{\infty} x^{-\alpha}dx \asymp cK^{1-\alpha}$ and $T_2(K) \asymp c^2 \int_K^{\infty} x^{-2\alpha}dx \asymp c^2K^{1-2\alpha}$, so $\sigma_K \asymp K^{-1/2}$.

If $1/2 < \alpha < 1$: $T_1(K) \asymp c\int_K^{p} x^{-\alpha}dx \asymp c\,(p^{1-\alpha} - K^{1-\alpha}) \asymp c\,p^{1-\alpha}$ and $T_2(K) \asymp c^2K^{1-2\alpha}$, giving $\sigma_K \asymp (cK^{(1-2\alpha)/2})/(cp^{1-\alpha}) = \frac{1}{\sqrt{K}}\left(\frac{K}{p}\right)^{1-\alpha}$. □

D.6 Theorem A.2: limits of association-only causal decisions

We now state a Bayes-error benchmark for the magnitude-only classification problem: how well can a rule that sees only the adjusted-association statistic T_{ij} classify a pair as causally linked versus correlated by background dependence under a stylized mixture? The result is a benchmark for what magnitude alone delivers when signal and background overlap by construction. It is *not* an impossibility theorem for inference about ρ_{ij}, which improves with n.

Theorem 3 (Association-only decisions cannot be reliable below the crud scale)**.** *Fix K and consider a uniformly random feature pair (i, j) with $i \neq j$. Suppose the analyst compresses the data to a one-dimensional adjusted association statistic T_{ij} (for example $T_{ij} = \hat{\rho}_{ij}^{(K)}$) and uses a decision rule $\delta : \mathbb{R} \to \{0, 1\}$ where $\delta(T_{ij}) = 1$ means "direct causal edge present".*

Assume a stylized mixture model:

$$H_0 \text{ (no direct edge)}: \quad T_{ij} \sim N(0, \sigma_K^2), \tag{31}$$

$$H_1 \text{ (direct edge)}: \quad T_{ij} \sim N(\mu, \sigma_K^2), \tag{32}$$

where σ_K is the crud scale from Theorem 2 and μ is the mean shift induced by the direct causal effect after generic adjustment.

Then (for equal prior probabilities on H_0 and H_1) the minimum achievable misclassification error is

$$\inf_{\delta} \ \mathbb{P}\big(\delta(T_{ij}) \text{ is wrong}\big) = \Phi\left(-\frac{|\mu|}{2\sigma_K}\right). \tag{33}$$

In particular, if $|\mu| \leq c\,\sigma_K$, then

$$\inf_{\delta} \ \mathbb{P}\big(\delta(T_{ij}) \text{ is wrong}\big) \ \geq \ \Phi(-c/2), \tag{34}$$

a constant bounded away from 0 *independent of sample size.*

Proof. Under the stated model, T_{ij} follows one of two normal distributions with equal variance: $N(0, \sigma_K^2)$ under H_0 and $N(\mu, \sigma_K^2)$ under H_1. By the Neyman–Pearson lemma, the likelihood ratio test is optimal. For equal priors and $\mu > 0$, the optimal threshold is $t^\star = \mu/2$ (symmetrically $-\mu/2$ for $\mu < 0$). Then

$$\mathbb{P}(\text{error} \mid H_0) = \mathbb{P}(T_{ij} > \mu/2 \mid H_0) = 1 - \Phi\left(\frac{\mu}{2\sigma_K}\right), \tag{35}$$

$$\mathbb{P}(\text{error} \mid H_1) = \mathbb{P}(T_{ij} \leq \mu/2 \mid H_1) = \Phi\left(-\frac{\mu}{2\sigma_K}\right). \tag{36}$$

These are equal, so the Bayes error equals $\Phi(-\mu/(2\sigma_K))$. Replacing μ by $|\mu|$ handles both signs. The second inequality follows immediately if $|\mu| \leq c\sigma_K$. □

Remark (the bound is a benchmark, not always a lower bound). The equal-variance Gaussian model provides the cleanest impossibility benchmark. If additional structure such as variance changes is also present under H_1, identification may become easier (if signal pairs have distinctive variance) or harder (if the variance change increases overlap). The mean-shift formulation isolates the most basic obstruction: when the causal signal in adjusted-correlation units is comparable to the background scale, no thresholding rule on T_{ij} alone can reliably separate signal from noise.

Remark (rare edges make the problem worse). If the prior probability of a direct causal edge is $\pi \ll 1$, then even when $|\mu|/\sigma_K$ is moderate, the positive predictive value of any thresholding rule can remain small. Theorem 3 isolates the more basic point: when $|\mu|$ is on the order of the crud scale, no association-only rule can achieve low error.

Remark (robustness of the limit). The Gaussian mixture model is a stylized proxy for association-only workflows that compress each pair to a one-dimensional statistic. The qualitative limit depends on signal-to-noise: if the causal signal is not large relative to the background scale, discrimination is poor. Unequal variances or heavier-tailed distributions only make separation harder, and any rule that discards multivariate structure is subject to the same basic limitation.

D.7 Robustness of the assumptions

The theorems above use idealized assumptions—Haar-like eigenvectors, diagonal concentration, and a Gaussian mixture—to obtain closed-form expressions. Real datasets exhibit moderate eigenvector localization, finite-p effects, and heavy-tailed residuals. These deviations mainly affect constants and tail behavior, not the qualitative conclusions. When eigenvectors are partially localized, the spectrum-only formula in Theorem 2 typically becomes conservative (localization increases diagonal heterogeneity, reducing average off-diagonal energy). The Gaussian error curve in Theorem 3 is likewise a benchmark: heavy-tailed backgrounds only increase overlap for a fixed mean shift. Appendix E audits these assumptions empirically across all nine datasets: the spectral prediction tracks the observed residual SD with $r > 0.94$ in seven of nine datasets (weaker for CIFAR-10 and RNA-Seq), eigenvector coherence runs 3.8–11.2× the Haar ideal, and residual correlations are heavier-tailed than Gaussian. Each of these deviations biases the closed-form expressions in a known direction (mostly conservative for our claims, optimistic for the Gaussian error in Theorem 3), so the qualitative conclusions are not overturned, but they are not free of approximation.

E Empirical audit of assumptions

Theorems 2 and 3 rely on idealized assumptions; the question is whether these assumptions hold well enough in practice to support the qualitative conclusions. We test this on nine real-world datasets spanning diverse domains: Kay fMRI (visual neuroscience, n=8428, p=1870), Haxby fMRI (cognitive neuroscience, n=23612, p=1452), Stringer neural recordings (mouse cortex, n=7018, p=6000), NHANES (population health, n=29902, p=165), CIFAR-10 images (n=30000, p=3072), precinct-level voting and census demographics (n=28934, p=83), RNA-Seq gene expression (Allen Institute mouse brain, n=10071, p=387), GTEx gene expression (human skeletal muscle, n=10000, p=803), and HEXACO personality inventory (n=22786, p=242). All datasets are z-scored by feature before analysis; up to n=10000 samples are used per dataset. PCA is performed via randomized SVD with up to 300 components. For each test, we report both a diagnostic summary and an interpretation of what deviations from the ideal mean for the qualitative message.

E.1 Test 1: Power-law eigenvalue decay

Theorem 2 and Corollary 1 assume that eigenvalues follow an approximate power law $\lambda_k \propto k^{-\alpha}$. We fit $\log(\lambda_k/\text{total var}) = a - \alpha \log k$ on the first 40 components and compare to an exponential model via AIC.

Table E.1: Power-law fits on first 40 principal components.

Dataset	α	R^2	AIC (power law)	AIC (exponential)	Preferred
NHANES	0.63	0.992	−237	−121	power law
Kay fMRI	0.67	0.984	−204	−97	power law
Precinct	0.70	0.985	−205	−105	power law
Stringer	0.74	0.987	−206	−109	power law
HEXACO	0.97	0.974	−155	−70	power law
Haxby fMRI	1.03	0.982	−165	−98	power law
CIFAR-10	1.24	0.997	−218	−64	power law
GTEx	1.30	0.994	−190	−62	power law
RNA-Seq	1.33	0.932	−90	−24	power law

All nine datasets are better described by a power law than an exponential over the first 40 components (ΔAIC > 60 in every case). Exponents range from $\alpha \approx 0.63$ (NHANES) to $\alpha \approx 1.33$ (RNA-Seq), placing all datasets in the regime $\alpha > 1/2$ required by Corollary 1. Most datasets cluster near $\alpha \approx 0.7$–1.0, consistent with $1/f$-like spectral decay.

E.2 Test 2: Eigenvector delocalization

Assumption 1 requires that eigenvectors are not concentrated on small subsets of coordinates. We measure this via coherence $\mu := \max_{i,k} |v_{ik}|$ (ideal: $1/\sqrt{p}$ under Haar), the inverse participation ratio $\text{IPR}(k) := \sum_i v_{ik}^4$ (ideal: $1/p$), and leverage score uniformity.

The degree of delocalization varies systematically with p. High-dimensional datasets (Stringer, CIFAR-10, GTEx, Kay, Haxby) show moderate delocalization with coherence ratios of 3.8–6.4× ideal and effective supports spanning 25–50% of p. Low-dimensional datasets (Precinct, NHANES) exhibit substantial localization, with coherence ratios up to 8.3× and effective supports well below p. RNA-Seq shows the most localization (11.2×), consistent with the presence of highly co-regulated

Table E.2: Eigenvector delocalization diagnostics (top 50 PCs).

Dataset	Coherence	Ideal $1/\sqrt{p}$	Ratio	Med. eff. support	p
Stringer	0.082	0.013	6.4×	1542	6000
GTEx	0.194	0.035	5.5×	245	803
CIFAR-10	0.069	0.018	3.8×	1216	3072
Kay fMRI	0.118	0.023	5.1×	556	1870
Haxby fMRI	0.131	0.026	5.0×	478	1452
RNA-Seq	0.572	0.051	11.2×	57	387
HEXACO	0.371	0.064	5.8×	73	242
NHANES	0.439	0.078	5.6×	29	165
Precinct	0.908	0.110	8.3×	12	83

gene modules. As discussed in Section D.7, localization typically makes the spectrum-only formula *conservative* (overestimates typical correlations), so the qualitative conclusions of Theorem 2 remain valid even where eigenvectors are not perfectly delocalized.

E.3 Test 3: Predicted σ_K versus empirical residual correlation SD

The central prediction of Theorem 2 is that the standard deviation of off-diagonal residual correlations after removing K PCs is approximately $\sigma_K = \sqrt{T_2(K)}/T_1(K)$. We compare this spectral prediction to the empirically measured SD of residual correlations on a random subset of 500 features.

Table E.3: Correlation between predicted σ_K and empirical residual correlation SD across $K \in \{0, 1, 2, 5, 10, 20, 50, 100\}$.

Dataset	Pearson r (predicted vs. empirical)
HEXACO	0.997
Precinct	0.997
Haxby fMRI	0.996
NHANES	0.987
GTEx	0.987
Stringer	0.952
Kay fMRI	0.942
CIFAR-10	0.880
RNA-Seq	0.639

In seven of nine datasets, the spectral prediction tracks the empirical SD with $r > 0.94$. CIFAR-10 ($r = 0.88$) and RNA-Seq ($r = 0.64$) show weaker agreement, the former due to strong eigenvector structure (spatial image statistics) and the latter due to gene-module localization. Even where the spectral formula is quantitatively imprecise, the monotonic relationship between K and the crud scale is preserved, supporting the qualitative message that broad spectra produce slowly decaying background correlations.

E.4 Test 4: Diagonal concentration of residual covariance

The proof of Theorem 2 requires that the diagonal entries $\Sigma_{ii}^{(K)}$ concentrate around their mean $T_1(K)/p$. We measure this via the coefficient of variation (CV) of the residual variance $\mathrm{Var}_n(X_i^{(K)})$ across features i.

Table E.4: Coefficient of variation of residual variances across features.

Dataset	K=0	K=1	K=10	K=50
Stringer	0.000	0.036	0.091	0.136
CIFAR-10	0.000	0.217	0.222	0.216
Kay fMRI	0.000	0.052	0.105	0.167
Haxby fMRI	0.000	0.122	0.149	0.163
HEXACO	0.000	0.097	0.171	0.220
GTEx	0.000	0.303	0.516	0.689
NHANES	0.177	0.234	0.456	0.811
RNA-Seq	0.000	0.550	0.544	0.320
Precinct	0.000	0.177	0.589	1.127

The pattern is clear: diagonal concentration holds well for large p and degrades predictably as p shrinks and K grows. At K=0 (before PC removal), z-scoring enforces unit variance by construction (CV ≈ 0). As PCs are removed, diagonal heterogeneity increases. High-p datasets (Stringer, Kay, Haxby) maintain good concentration (CV < 0.2) even at K=50. Low-p datasets (Precinct, NHANES) show substantial heterogeneity at large K, consistent with the finite-p discussion in Section D.7. This diagonal heterogeneity makes the Haar-based formula conservative, so the spectrum-only prediction is an upper bound on typical off-diagonal correlations in these settings.

E.5 Test 5: Normality of residual correlations

Theorem 3 models the null distribution of residual associations as Gaussian. We assess this by computing the skewness and excess kurtosis of the off-diagonal residual correlations.

Table E.5: Skewness and excess kurtosis of off-diagonal residual correlations.

	K=0		K=10	
Dataset	Skew	Kurt	Skew	Kurt
HEXACO	0.07	1.94	0.26	17.36
GTEx	−0.26	−0.35	0.30	1.41
Haxby fMRI	0.33	−0.34	0.17	1.34
Kay fMRI	0.39	5.46	1.45	24.07
CIFAR-10	0.99	0.63	2.47	13.99
Precinct	1.25	8.44	0.86	8.87
RNA-Seq	−1.61	1.96	1.98	12.61
Stringer	1.98	14.49	1.23	11.00
NHANES	2.99	21.43	2.90	35.58

Residual correlations are generally right-skewed with positive excess kurtosis (heavier tails than Gaussian). GTEx and Haxby fMRI at K=0 are the closest to Gaussian (skew −0.26/0.33, kurtosis

$-0.35/-0.34$). NHANES shows the largest departures. As noted in Section D.7, heavy tails increase the overlap between null and alternative distributions for a given mean shift, so the Gaussian error formula in Theorem 3 is optimistic about separability—the true error is at least as large.

E.6 Test 6: Cross-component covariance negligibility

Assumption 1 requires that the covariance of $v_{ik}v_{jk}$ and $v_{i\ell}v_{j\ell}$ for $k \neq \ell$ is negligible. We estimate this by computing the lag-1 autocorrelation of the product sequence $\{v_{ik}v_{jk}\}_{k=1}^{K}$ across 1000 random variable pairs (i, j).

Table E.6: Cross-component covariance: autocorrelation of eigenvector products.

Dataset	Mean cross-corr	|mean|	Frac. $\|r\| > 0.1$
RNA-Seq	-0.013	0.013	0.493
CIFAR-10	-0.018	0.018	0.469
Kay fMRI	-0.019	0.019	0.467
GTEx	-0.027	0.027	0.483
Stringer	-0.025	0.025	0.466
Haxby fMRI	-0.024	0.024	0.459
HEXACO	-0.028	0.028	0.497
NHANES	-0.035	0.035	0.552
Precinct	-0.061	0.061	0.568

The mean cross-component correlation is small in magnitude (< 0.07) for all datasets, with higher-p datasets showing smaller values as expected. While roughly half of individual pairs show $|r| > 0.1$, these are centered near zero and largely cancel in the variance calculation, consistent with the $O(p^{-3})$ bound in Assumption 1.

E.7 Summary of assumption audit

The assumptions of Theorems 2 and 3 hold to a reasonable approximation across all nine datasets, with the quality of the approximation improving systematically with p. The key findings are:

- Power-law eigenvalue decay ($\alpha \in [0.63, 1.33]$, $R^2 > 0.93$) is universally preferred over exponential decay.
- Eigenvector delocalization is moderate in high-p datasets and degrades gracefully in low-p settings.
- The spectral prediction σ_K tracks the empirical crud scale with $r > 0.88$ in eight of nine datasets.
- Diagonal concentration holds well for large p and degrades predictably for small p and large K.
- Residual correlations are heavier-tailed than Gaussian, making the error bound in Theorem 3 conservative (the true error is at least as large).
- Cross-component covariances are small in the mean, supporting the negligibility condition.

F Evidence that small-effect overclaiming is widespread

The main text argues that writing as if small adjusted associations were substantively meaningful is a widely shared practice across empirical sciences. This appendix gives a compact, field-by-field summary of the meta-research behind that claim. Each row of the table is anchored on a single canonical meta-research paper, and every quantitative claim in the row is taken directly from that paper, so each row can be audited against one source. Power/inflation and effect-size distributions are linked mechanistically: Gelman and Carlin [2014] show that significance filtering under low power inflates published effect magnitudes (Type M error), and Ioannidis [2008] documents the corresponding empirical pattern across disciplines. Table rows are ordered from the most observational fields (for which the crud-scale argument applies most directly) to the most design-mixed.

How this table was built. Fields were selected to span the empirical sciences where small adjusted observational associations are routinely interpreted as causal evidence. Within each field we anchor the row on a single widely-cited meta-research paper that quantifies either statistical power, effect-size inflation, or the distribution of published effect sizes; all numbers in the row come from that paper. We did not perform a de novo meta-analysis. For "Design mix", clean published experimental-vs-observational shares are not available for most fields; entries are a qualitative dominant-design label. All numbers are summary statistics of effect-size distributions, not single-study point estimates, and are intended as order-of-magnitude benchmarks.

Table F.1: Field-by-field evidence that small published effects are often underpowered and inflated, paired with the prevailing study-design mix and with the typical effect-size magnitudes that published work actually discusses. Each row is anchored on a single meta-research paper; every number in the row is taken verbatim from that paper, in the units the paper itself uses (RR, r, r^2, or fraction of genetic variance). Rows are ordered from most observational (top) to most design-mixed (bottom). The conversions between these units (Pearson $|r|$ from RR or from r^2) used in the surrounding discussion are derivations, not numbers from the cited papers. The crud-aware test complements power- and inflation-based diagnostics: even when a study is well-powered for its point null, a statistically significant adjusted association of typical field magnitude may still be indistinguishable from the domain's background dependence.

Field	Design mix	Reference	What the paper shows	Key number from that paper	Typical published effect size reported
Medicine / observational epidemiology	Predominantly observational	Siontis and Ioannidis [2011]	Survey of nominally significant risk-factor and intervention associations with tiny effects in the published medical literature.	51 nominally significant tiny-effect estimates ($p < 0.05$, RR $\in [0.95, 1.05]$) reported in 40 articles; in 22 of those 40 articles (55%) the authors expressed no concern about effect-size magnitude.	Reported RR is bounded by construction at $0.95 \leq \text{RR} \leq 1.05$.

continued on next page

Table F.1 – *continued from previous page*

Field	Design mix	Reference	What the paper shows	Key number from that paper	Typical published effect size reported
Nutritional epidemiology	Predominantly observational	Schoenfeld and Ioannidis [2013]	"Cookbook review" of cancer associations for 50 common ingredients drawn from a random recipe book.	40 of 50 ingredients had at least one cancer study; among those primary studies, 39% concluded increased cancer risk, 33% decreased risk, 5% borderline, and 23% no association.	Across the corresponding meta-analyses, RRs were on average null: median RR = 0.96, IQR 0.85–1.10.
Genetics (common-variant)	Observational by design	Park et al. [2010]	Direct estimation of the effect-size distribution for SNPs identified by GWAS, fitted to height, Crohn's disease, and breast/prostate/colorectal (BPC) cancers.	Per-variant effect sizes of the SNPs detected up to that point span 0.04–1.96% of genetic variance across the studied traits.	For height, ∼201 estimated susceptibility loci jointly explain ∼16% of genetic variance; the estimated underlying distribution implies many more loci with even smaller per-variant effects remain undiscovered.
Neuroscience (BWAS)	BWAS observational	Marek et al. [2022]	Brain-wide association studies in three large neuroimaging cohorts; tests the sample size needed for reliable detection.	Reliable detection of typical brain–behavior associations requires sample sizes in the thousands.	Median univariate brain–behavior $\lvert r\rvert \approx 0.01$; top 1% of associations reach $\lvert r\rvert > 0.06$; single largest replicated effect $\lvert r\rvert = 0.16$.
Psychology	Mixed (experiment-leaning)	Richard et al. [2003]	Century-scale quantitative synthesis of social-psychology effect sizes.	474 meta-analytic effects synthesized from >25,000 primary studies.	Mean $\lvert r\rvert \approx 0.21$ across all synthesized effects.
Economics	Predominantly observational	Ioannidis et al. [2017]	Power meta-analysis of the empirical economics literature, surveying 159 areas of research drawing on 64,076 estimates from over 6,700 studies.	Median statistical power is 18% or less; nearly 80% of adequately powered reported effects are exaggerated, typically by a factor of two.	Average effect size across the surveyed estimates (measured largely as partial correlations and elasticities) is 0.17.
Behavioural ecology / evolution	Mixed (lab + field)	Møller and Jennions [2002]	Synthesis of effect sizes across ecology and evolution meta-analyses.	43 published meta-analyses synthesized, comprising 93 mean-effect estimates using Pearson's r and 136 using Hedges' d or g.	Mean variance explained $r^2 = 2.51$–5.42% across the synthesized estimates (95% CIs 1.99–7.05%).

A consistent pattern emerges. The reported magnitudes are in different native units across rows, but standard conversions place them on a common scale. Pearson $|r|$ values are reported directly only by Marek et al. [2022] (median brain–behavior $|r| \approx 0.01$, single largest replicated $|r| = 0.16$), Richard et al. [2003] (mean $|r| \approx 0.21$), and Ioannidis et al. [2017] (average partial

correlation/elasticity ≈ 0.17). For the remaining rows, applying the binary RR-to-r conversion of Equation (16) (Appendix D.3) and the $|r| = \sqrt{r^2}$ identity gives: RR $\in [0.95, 1.05] \Rightarrow |r| \lesssim 0.025$ (medicine, evaluated at high outcome prevalence; at the low prevalences typical of rare-disease epidemiology, $|r|$ is several-fold smaller), median RR $= 0.96 \Rightarrow |r| \approx 0.02$ (nutrition, similarly prevalence-sensitive), per-variant variance $\leq 1.96\%$ of genetic variance combined with heritabilities of order 0.5–0.8 implying per-variant $|r| \lesssim 0.13$ (genetics), and $r^2 = 2.5$–$5.4\% \Rightarrow |r| \approx 0.16$–$0.23$ directly (behavioural ecology). All converted values inherit the caveats in Appendix D.3. Together the typical published-effect magnitudes across the seven fields cluster in the range $|r| \approx 0.01$–0.23. Critically, these typical effect sizes are of the same order as—and in several fields smaller than—the empirical crud scales we report across nine datasets (Table 1, where $\sigma_{\text{crud}}^{(K=0)}$ ranges from 0.04 to 0.19). The "Design mix" column also locates where the paper's contribution is sharpest: in the four fields that are essentially entirely observational (medicine / epidemiology, nutritional epidemiology, common-variant genetics, BWAS-style neuroscience), the crud-scale argument applies directly; in the three fields with substantial experimental or quasi-experimental components (psychology, economics, behavioural ecology), randomization or design-based identification breaks the generic-dependence problem for the experimental subset, but power and inflation issues persist. The interpretive step this paper targets—treating a small, statistically significant adjusted association as support for a causal claim—is therefore being taken under conditions where, on average, the cited meta-research finds power inadequate and reported effects inflated, and against a background of generic dependence that is numerically of similar magnitude. Our contribution is orthogonal and complementary to the power/inflation literature: we ask, in addition, whether a given small adjusted association is distinguishable from the domain's empirical null.

Would the typical published effect clear the crud-aware threshold? A direct way to read the table above against the test we propose is to ask, for each field, whether the typical reported magnitude exceeds the conventional 95th-percentile crud-aware threshold ($z_{\text{crud}} > 1.96$, i.e. $|r| > 2\sigma_K$) against an approximately matched reference σ_K from Table 1 at $K = 10$. The matchings below are illustrative, not domain-specific—the right comparison for a given paper would use a same-pipeline same-population σ_K.

Table F.2: Field-level signal-to-crud ratios. "Typical $|r|$" is taken from the converted values in the conclusion above; reference σ_K is matched approximately to each field's data modality at $K = 10$ (Table 1); "Verdict" is whether the typical effect clears $|r|/\sigma_K > 2$. Caveats from Appendix D.3 apply throughout.

Field	Typical $\lvert r\rvert$	Reference σ_K (dataset)	$\lvert r\rvert/\sigma_K$	Verdict
Medicine (RR$\sim$1.05)	0.025	0.093 (NHANES)	0.27	fails
Nutrition (median RR 0.96)	0.04	0.093 (NHANES)	0.43	fails
Genetics (per-variant)	$\lesssim 0.13$	0.071 (RNASeq)	$\lesssim 1.8$	fails / marginal
Neuroscience (BWAS median)	0.01	0.035 (Kay fMRI)	0.29	fails
Economics	0.17	0.093 (NHANES)	1.8	marginal
Beh. ecology	0.18–0.23	—	—	no analog

The pattern is striking: in observational medicine, nutritional epidemiology, common-variant genetics, and BWAS-style neuroscience, the typical published effect would *not* clear the crud-aware threshold against a matched reference. Economics lands near the threshold; behavioural ecology has no clean analog in our nine datasets. Psychology is omitted from the table: the row's natural

anchor (Richard 2003's mean $|r| \approx 0.21$) draws from meta-analyses with substantial experimental content, and we lack a meta-analytic synthesis of predominantly observational psychology of comparable scope to anchor the row. The HEXACO worked example in Section 2.8 provides the qualitative picture for the observational individual-differences subset (within-facet item pairs clear σ_K comfortably). The fields whose typical effects fall below the threshold are the same fields the meta-research literature has flagged for power and inflation problems—the test is not adding a new diagnosis, but it is providing a quantitative pass/fail at the per-paper level that the power/inflation literature has not.

G Robustness checks

The main text documents approximately power-law eigenvalue spectra across all datasets and argues that the resulting broad spectral structure drives slow decay of the background correlation scale under PC regression. Two natural concerns arise: (i) could the observed approximately power-law spectrum be an artifact of heavy tails, outliers, or scaling properties of the raw data rather than a genuine feature of the dependence structure? and (ii) could the spectral structure be consistent with independence among features, so that the observed spectra are explainable by noise alone? Both concerns can be addressed with systematic robustness checks, and in every case the power-law structure survives.

G.1 Heavy-tail stress test: is the power-law spectrum a preprocessing artifact?

Approximately $1/f$ structure could in principle arise from artifacts such as outlier entries, heterogeneous row scales, or heavy-tailed marginal distributions, rather than from genuine shared dependence among features. If any of these were the dominant driver, then transformations that remove the artifact should noticeably weaken or destroy the power-law behavior. We test this by applying four transformations to each dataset before recomputing the PCA spectrum:

1. **Clipping** at the 99.9th percentile: all entries beyond the 0.1%/99.9% quantiles are clipped to the boundary values, removing extreme outliers while preserving the bulk of the distribution.
2. **Row-wise L_2 normalization**: each row (observation) is divided by its Euclidean norm, removing heterogeneous magnitude scaling across samples.
3. **Rank Gaussianization**: all entries are replaced by their inverse-normal-transformed ranks (via $z = \sqrt{2}\,\mathrm{erfinv}(2u-1)$ where u is the rank-based uniform quantile), completely removing marginal non-Gaussianity while preserving the rank-order dependence structure.
4. **Gaussian null**: an i.i.d. $N(0,1)$ matrix of matching dimensions (capped at 5000×2000) provides a baseline with no dependence structure.

For each transformation, we fit the power law $\log(\lambda_k/\text{total var}) = a - \alpha \log k$ on the first 40 components (matching the procedure in Section E) and report the exponent α and goodness of fit R^2. We also compute the Hill tail index $\hat{\mu}$ on the top 2% of absolute entry values, both on the raw data and after row normalization, to characterize marginal tail heaviness.

The results (Table G.1) show that the power-law spectral structure is robust to all transformations. Clipping at the 99.9th percentile has essentially no effect on either α or R^2 in any dataset, ruling out extreme outlier entries as the driver. Row normalization and rank Gaussianization produce modest shifts in α (typically < 0.15) but preserve the qualitative power-law shape with

Table G.1: Heavy-tail stress test: power-law fits (α, R^2) on the first 40 principal components under different data transformations. The Gaussian null (i.i.d. noise) produces a flat spectrum with no power-law structure.

	Original		Clip@0.999		Row-norm		Rank-Gauss	
Dataset	α	R^2	α	R^2	α	R^2	α	R^2
Kay fMRI	0.67	0.98	0.67	0.98	0.67	0.98	0.67	0.98
Haxby fMRI	1.03	0.98	1.02	0.98	0.87	0.94	0.97	0.96
NHANES	0.63	0.99	0.64	0.99	0.91	0.93	0.83	0.99
CIFAR-10	1.24	1.00	1.24	1.00	1.09	0.99	1.25	1.00
GTEx	1.30	0.99	1.30	0.99	1.24	0.99	1.30	0.99
Precinct	0.70	0.99	0.70	0.99	0.76	0.97	0.77	0.98
RNA-Seq	1.33	0.93	1.33	0.93	1.02	0.94	1.44	0.93
HEXACO	0.97	0.97	0.97	0.97	0.94	0.97	1.02	0.97
Stringer	0.74	0.99	0.74	0.99	0.72	0.99	0.89	0.99

high R^2 (> 0.93 in all cases). The most notable change is that rank Gaussianization can slightly increase α (e.g., RNA-Seq: $1.33 \to 1.44$; Stringer: $0.74 \to 0.89$), suggesting that heavy tails in marginals, if anything, slightly flatten the spectrum rather than create the power law. By contrast, the Gaussian null (i.i.d. noise of matching dimensions) produces a flat spectrum with no power-law structure, confirming that the observed spectral decay reflects genuine shared dependence rather than a finite-sample artifact.

These checks support treating the broad, approximately $1/f$ eigenvalue spectrum as a real structural property of these datasets. The power-law behavior persists whether or not entries are clipped, rows are rescaled, or marginals are Gaussianized, and it is absent in data with no dependence structure.

G.2 Null-model comparisons: are the spectra consistent with independence?

A complementary question is whether the observed eigenvalue spectra could arise from independent features. If features were truly independent (or had only marginal structure but no cross-feature dependence), then the PCA spectrum should resemble that of a noise matrix. We compare the empirical spectrum of each dataset against two null models:

1. **Gaussian null**: an i.i.d. $N(0, 1)$ matrix of matching dimensions ($n \times p$, capped at 5000×2000). Under the Marchenko–Pastur law [Marčenko and Pastur, 1967], eigenvalues cluster in a bounded interval with no power-law tail.

2. **Column-permutation null**: the values within each column are independently shuffled across rows, destroying all cross-feature correlations while preserving each feature's marginal distribution exactly. This is a stricter control than the Gaussian null because it retains marginal non-Gaussianity, sparsity, and any column-specific scaling. We average over 3 independent permutation realizations.

For each null model, we compute both the explained-variance spectrum and the inverse participation ratio ($\mathrm{IPR}_k = \sum_i v_{ik}^4$) of each eigenvector, which measures eigenvector localization.

Results. The key question is whether the observed spectra could arise from independent features. Across all nine datasets, the empirical eigenvalue spectrum separates dramatically from both null

models. The empirical spectra show power-law decay spanning 1–2 orders of magnitude above the null baseline, while both the Gaussian and column-permutation nulls produce flat spectra clustered near $1/p$. The separation is evident from the first principal component onward and persists across the full range of computed components (up to 300). The broad spectral structure reflects genuine cross-feature dependence rather than marginal properties of individual features.

The column-permutation null is particularly informative: it preserves each feature's exact marginal distribution (including heavy tails, discreteness, and sparsity) but destroys all pairwise and higher-order dependence. The fact that the column-permutation spectrum is indistinguishable from the Gaussian null confirms that the observed power-law structure arises from cross-feature dependence, not from marginal distributional properties.

Eigenvector localization provides a complementary diagnostic. Empirical eigenvectors show moderately elevated IPR values compared to the null models, particularly for the leading components, consistent with the partial eigenvector localization documented in Section E (Test 2). However, the IPR elevation is modest (typically 1.5–3× the null level) and diminishes for higher-rank components. Eigenvectors are not perfectly delocalized, but their localization is far from extreme enough to invalidate the qualitative conclusions.

G.3 Summary

The robustness checks confirm two key properties of the datasets studied:

- The approximately power-law eigenvalue spectrum is not an artifact of outliers, row-scale heterogeneity, or marginal non-Gaussianity. It persists under clipping, row normalization, and rank Gaussianization, and it is absent in independence-preserving null models (Tables G.1).
- The spectral structure reflects genuine cross-feature dependence: column permutation (which preserves marginals but destroys dependence) eliminates the power-law decay entirely.

Together with the assumption audit in Section E, these checks support the main text's interpretation: the broad spectral structure that drives slow decay of the crud scale under PC regression is a real feature of these datasets' dependence structure, not a preprocessing or distributional artifact.

G.4 Alternative adjustment methods: the problem is the spectrum, not PCA

The main text uses top-K PC regression as a deliberately generic adjustment strategy. A natural concern is whether the persistence of broad residual correlations is specific to PCA, or whether it reflects a more fundamental property of the datasets' dependence structure that would limit any moderate-dimensional linear adjustment. We test this by repeating the core analysis—adjust all features, recompute pairwise correlations on residuals, measure the spread—under three alternative adjustment strategies that are commonly used or proposed in observational causal inference practice.

Methods.

1. **Random covariate regression.** For each feature, regress it on a randomly chosen set of m other features (via OLS) and take the residual. This mimics the common practice of "controlling for observables" when no principled adjustment set is available. We average over 5 independent random covariate draws.

2. **Random projection adjustment.** Project the data onto K random Gaussian directions (orthonormalized), regress those out, and measure residual correlation spread. This is a useful control because it tests whether the *choice* of directions matters: if random projections give similar results to PCA, the problem is the dimensionality of shared variation, not the specific directions removed. We average over 5 independent random projection draws.

3. **Lasso adjustment.** For each feature in the correlation subset, use Lasso (ℓ_1-penalized regression, $\alpha = 0.1$) to data-adaptively select predictors from all other features, then compute pairwise correlations on the residuals. This tests whether data-driven variable selection can isolate the "right" confounders and collapse the background dependence. It maps onto doubly-robust and post-double-selection workflows popular in econometrics [Belloni et al., 2014].

For comparability, we measure the same quantity across all methods: the standard deviation of off-diagonal residual correlations on a fixed random subset of 300 features (or all features if $p < 300$), at adjustment dimensions $m \in \{10, 50, 100\}$ for the linear methods.

Table G.2: Residual correlation SD under alternative adjustment strategies. PCA targets the top eigenvalues; random covariates mimic observational practice; random projections remove generic directions; Lasso uses data-driven variable selection. All methods leave substantial residual spread.

	Dimension 10			Dimension 50			
Dataset	PCA	Rand. cov.	Rand. proj.	PCA	Rand. cov.	Rand. proj.	Lasso
Kay fMRI	0.036	0.059	0.065	0.027	0.044	0.065	0.032
Stringer	0.019	0.028	0.030	0.015	0.024	0.030	0.017
HEXACO	0.046	0.099	0.145	0.040	0.076	0.146	0.033
Haxby fMRI	0.079	0.129	0.177	0.036	0.075	0.176	0.042
RNA-Seq	0.071	0.125	0.182	0.043	0.067	0.227	0.037
GTEx	0.101	0.170	0.285	0.056	0.095	0.284	0.063
CIFAR-10	0.104	0.139	0.190	0.062	0.090	0.193	0.084
NHANES	0.093	0.097	0.119	0.116	0.105	0.127	0.077
Precinct	0.136	0.135	0.169	0.309†	0.135	0.207	0.094

†Precinct has only p=83 features. Removing 50 of 83 PCs leaves a near-degenerate residual subspace, inflating the residual correlation SD. This is a finite-p artifact, not a property of the adjustment method.

Results (Table G.2). The qualitative picture is the same across all methods—all leave substantial residual spread—but the quantitative differences are real and informative.

1. **All methods leave substantial residual spread.** At dimension 10, the residual correlation SD ranges from 0.019 to 0.285 depending on dataset and method; no method collapses it to near zero. Even Lasso, which can draw on all $p - 1$ remaining features as potential predictors (not just K directions), produces residual SDs of 0.017–0.094.

2. **PCA is the most efficient per dimension; random covariates are 1.5–3× worse.** PCA targets the top eigenvalues by construction, so it removes the most shared variance per dimension. Random covariate regression—the procedure closest to common observational practice—produces residual SDs that are typically 1.5–2.2× larger than PCA at the same dimension (e.g., Haxby at dimension 10: PCA 0.079 vs. random covariates 0.129; HEXACO: 0.046 vs. 0.099). This gap reflects the fact that randomly chosen covariates are unlikely to align with the dominant shared directions, so they remove less dependence structure per covariate.

3. **Random projections are nearly inert for high-p datasets.** For CIFAR-10 (p=3072), Haxby (p=1452), and Stringer (p=6000), removing 10, 50, or 100 random directions produces virtually identical residual SDs (e.g., CIFAR-10: 0.190, 0.193, 0.194). This is the cleanest evidence for the spectral explanation: when the eigenvalue spectrum is broad, random directions miss the shared variance entirely, so projecting them out has negligible effect on the correlation structure.

4. **Lasso can outperform PCA but still cannot collapse the background.** Lasso uses data-driven variable selection from all other features, giving it access to more effective predictors than a fixed K-dimensional subspace. It often produces lower residual SDs than PCA at comparable effective dimensionality (e.g., Haxby: Lasso 0.042 vs. PCA at K=10: 0.079; Precinct: 0.094 vs. 0.136). Nevertheless, even Lasso leaves residual SDs of 0.017–0.094 across datasets, confirming that the background dependence cannot be eliminated by any single-step linear adjustment.

The methods differ by factors of roughly 1.5–3×, which is not negligible—but the central point is that none of them collapse the residual correlation spread to near zero. Whether one uses PCA, random covariates, Lasso, or random projections, residual correlation SDs remain in the range 0.02–0.28. The crud scale is a property of the data's spectral structure, not of the particular adjustment method.

G.5 Additional worked examples

NHANES biomarkers. Hemoglobin and hematocrit are known to be near-deterministically related (hematocrit $\approx 3\times$ hemoglobin by definition). In the NHANES dataset ($n = 29{,}902$), their correlation is $r = 0.91$. Applying the crud-aware z-test with $\sigma_{\text{crud}} = 0.093$ (NHANES at $K = 10$) gives $z_{\text{crud}} = 9.8$, $p_{\text{crud}} < 10^{-20}$. Similarly, ALT and AST (both liver transaminases reflecting hepatocyte damage) correlate at $r = 0.83$, giving $z_{\text{crud}} = 8.9$. Both associations survive the crud-aware test by a wide margin, as expected for pairs with known mechanistic relationships.

GTEx RNA-seq. Most cis-eQTLs reported in the GTEx v8 atlas have small effect sizes, with only about 22% showing >2-fold effects [The GTEx Consortium, 2020]. Using the GTEx v8 per-sample TPM data for skeletal muscle (top 10,000 genes by variance across 803 samples), the empirical null at $K = 10$ has a background SD of 0.10. A typical small cis-eQTL effect of $r = 0.10$ sits at only the 71st crud percentile—squarely within the bulk of background gene–gene correlations. Even $r = 0.20$ reaches only the 95th percentile. Only effects with $r > 0.30$ (the 99th percentile, 3× the background SD) are clearly distinctive. Since most reported cis-eQTLs have small effect sizes, much of the GTEx catalog may not be distinguishable from background transcriptomic dependence using association magnitude alone.

G.6 Positive control: known-strong pairs clear the empirical null

The empirical null framework (Section D.2) is designed to flag associations that stand out from the background. A natural concern is whether genuinely strong associations—pairs with known biological or psychometric relationships—do in fact survive this test. If the crud-aware null swallowed real signal, it would be too conservative to be useful.

We test this using two datasets where ground-truth group structure is available:

- **HEXACO personality inventory** (p=242): items within the same personality facet (e.g., the 10 Honesty-Sincerity items) are designed by psychometricians to measure the same underlying trait. There are 24 facets yielding 1080 within-facet pairs.
- **NHANES biomedical survey** (p=165): features within the same clinical subsystem (e.g., lipid panel: total cholesterol, HDL, triglycerides, LDL; or hematology: RBC, hemoglobin, hematocrit, MCV) are known to be biologically linked. Seven biomarker groups yield 54 within-group pairs.

For each dataset, we compute the residual correlation matrix after removing K PCs, then compute the crud-aware p-value for each known-strong pair: the fraction of all $\binom{p}{2}$ pairs with $|\hat{\rho}^{(K)}| \geq |\hat{\rho}_{ij}^{(K)}|$. If the positive control works, known-strong pairs should be heavily enriched in the extreme tails.

Table G.3: Positive control: known-strong pairs are enriched in the tails of the crud-aware null. "Top 5%" and "Top 1%" report the fraction of known-strong pairs with crud-aware p-value below 0.05 and 0.01, respectively (expected: 5% and 1% for random pairs).

Dataset	K	Median $\|r\|$	Median p_{crud}	Top 5%	Top 1%	BG SD
HEXACO	0	0.324	0.041	55%	20%	0.145
HEXACO	10	0.044	0.254	29%	15%	0.046
HEXACO	20	0.048	0.148	33%	14%	0.038
NHANES	0	0.389	0.016	69%	41%	0.115
NHANES	10	0.131	0.085	44%	30%	0.093
NHANES	20	0.390	0.010	70%	54%	0.099

Results (Table G.3). The central question is whether the crud-aware null is too aggressive—whether it swallows real signal along with background noise. Known-strong pairs are massively enriched in the tails of the crud-aware null at every level of adjustment, confirming that it does not.

In HEXACO, 55% of within-facet pairs fall in the top 5% at K=0, and 20% fall in the top 1%—enrichments of 11× and 20× over chance. After removing 10 PCs, 29% remain in the top 5% and 15% in the top 1% (6× and 15× enrichment). After 20 PCs, enrichment is similar. The median within-facet $|r|$ drops from 0.32 to 0.04–0.05 after adjustment (comparable to the background SD of 0.04–0.05), yet the enrichment persists because within-facet correlations are systematically larger than typical random-pair correlations at the same adjustment level. Psychometrically designed pairs clear the empirical null at every adjustment level.

In NHANES, the enrichment is even stronger: 69% of biomarker-group pairs are in the top 5% at K=0, rising to 70% at K=20 (where 54% are in the top 1%). This reflects the strong mechanistic links within clinical subsystems (e.g., hemoglobin and hematocrit are physiologically coupled, systolic and diastolic blood pressure are mechanically linked). Biologically grounded pairs are clearly distinctive against the domain background.

The upshot is that the empirical null framework separates signal from noise in the right direction: genuinely strong associations clear the empirical null and are clearly distinguished from the background, even after generic PC adjustment. At the same time, the bulk of random-pair associations remain near the crud scale, reinforcing the main text's conclusion that most observed associations in these domains reflect background dependence rather than specific causal links.

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